OVERVIEW Tailoring, Models, Methods, and Artifacts

Two parts. Tailoring is the deliberate adaptation of the project-management approach, governance and processes to the environment and the work: what can be tailored (life cycle, processes, engagement, tools, methods and artifacts), why (commitment, customer focus, efficient resources), and the four-step process (select an initial approach, tailor for the organization, tailor for the project, keep improving). The second part is the PMBOK toolbox: models (ways of thinking, such as situational leadership, Herzberg, ADKAR, Cynefin, Tuckman), methods (ways of doing, such as estimating and meetings) and artifacts (things produced, such as the business case, registers and plans).

Tailoring

Deliberate, continuous adaptation; more critical projects need more rigor. Steps: select → organization → project → improve.

What you tailor

Life cycle and approach, processes (add, modify, remove, blend, align), engagement, tools, methods and artifacts.

Model, method, artifact

Model = a way of thinking; method = a way of doing; artifact = a template, document or deliverable.

Models to know

Situational Leadership II, Herzberg's hygiene and motivators, Maslow, Theory X/Y/Z, ADKAR, Kotter's 8 steps, Cynefin, Stacey, Tuckman, salience.

HOW THE LECTURE IS ORGANIZED

Slides 1–12 Tailoring: definition, competing demands, why and what to tailor

Slides 13–22 The tailoring process: organization and project factors, ongoing improvement

Slides 23–27 Models, methods and artifacts: definitions and costs

Slides 28–34 Commonly used models

Slides 35–41 Commonly used methods and artifacts

EXAM FOCUS

The instructor said to understand these models rather than memorize them; expect scenarios such as "which model helps here?" or "is this a hygiene factor or a motivator?". Tuckman (Midterm 251 MCQ 7) is tested in Chapter 3, and Final 251 asked for a PM tool or technique for a complex software project.

HOW TO READ THIS PAGE

Every original slide is shown unchanged. Yellow marks on a slide point to a numbered note: hover or tap a mark to jump to its note, and hover a note to see its marks. Questions tagged Past exam appeared on a real paper; Prediction questions are likely angles, not confirmed. Pick an answer to each question: a wrong pick tells you why it is wrong, so you can try again.

SLIDE 1 / 41 **Slide 1**



Tailoring, Models, Methods, and Artifacts

SE423: Software Project Management

1

NOTES

1. **Tailoring** means fitting the project-management approach to the project. **Models, methods and artifacts** are the toolbox it picks from.
2. This finishes Chapter 1: principles (how to behave) and performance domains (what to manage) are applied through a tailored selection of tools.

SLIDE 2 / 41 **Outline**

Outline

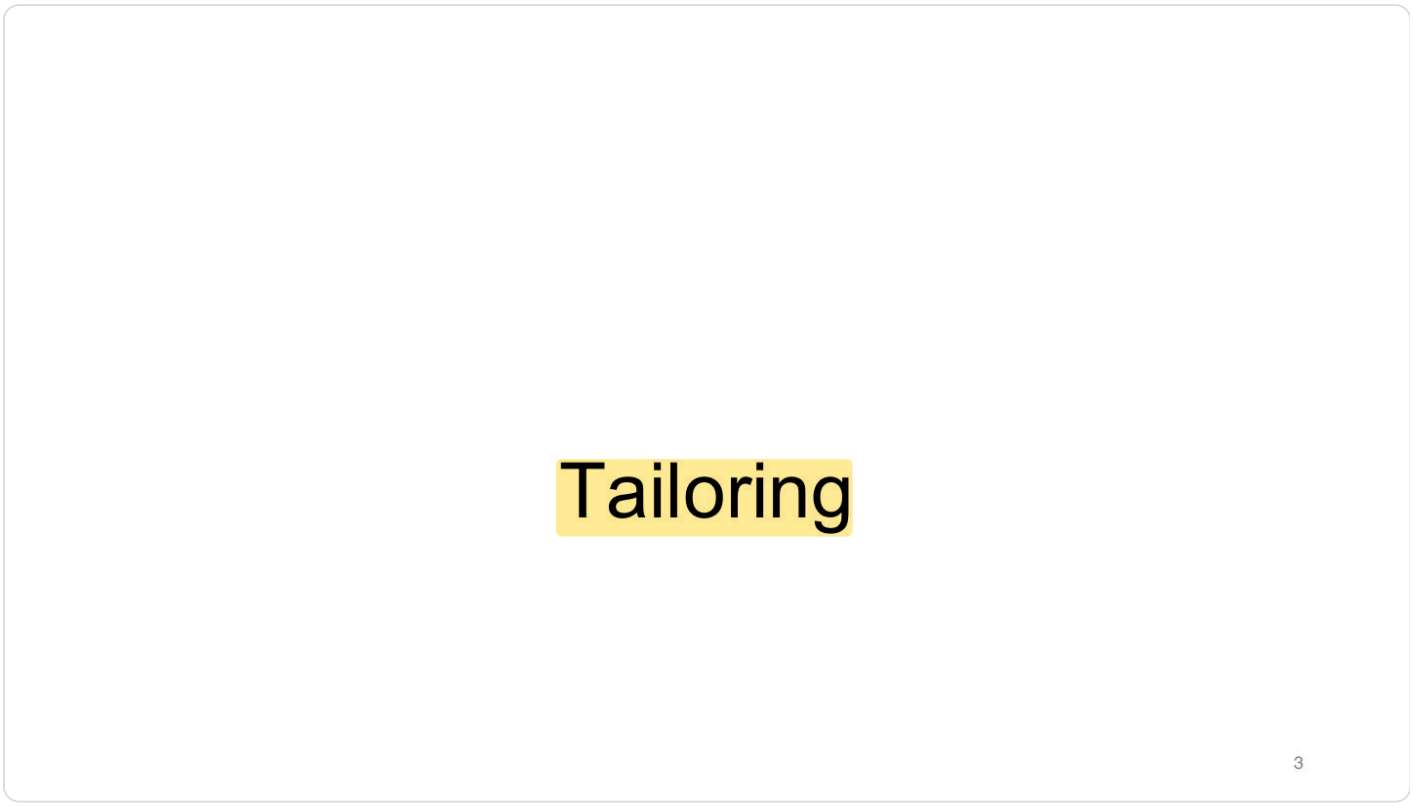
- Tailoring
- Models, Methods, and Artifacts

2

NOTES

1. Part 1, slides 3–22: tailoring.
2. Part 2, slides 23–41: models, methods and artifacts.

SLIDE 3 / 41 Tailoring (section)



NOTES

1. Section divider: tailoring.

Tailoring

Tailoring

- Tailoring is the deliberate adaptation of the project management approach, governance, and processes to make them more suitable for the given environment and the work at hand.
- In a project environment, tailoring considers the development approach, processes, project life cycle, deliverables, and choice of people with whom to engage.
- Many variables factor into the tailoring process, including the criticality of the project and the number of stakeholders involved.
- Using these variables as an example, it is evident that the rigor, checks and balances, and reporting required for a critical project (e.g., building a nuclear reactor) are much greater than those for building a new office building.

NOTES

1. **Tailoring** is the **deliberate** adaptation of the project-management approach, governance and processes to suit the environment and the work at hand. Deliberate is the key word: it is a conscious design choice, not cutting corners.
2. It covers the **development approach**, processes, project life cycle, deliverables and who to engage.
3. Many variables drive it, including the project's **criticality** and the number of stakeholders.
4. Example: a **nuclear reactor** needs far more rigor, checks and balances and reporting than a new office building. More critical means more process, never less.

EXAM ANGLE

Prediction Q1

Which statement best defines tailoring?

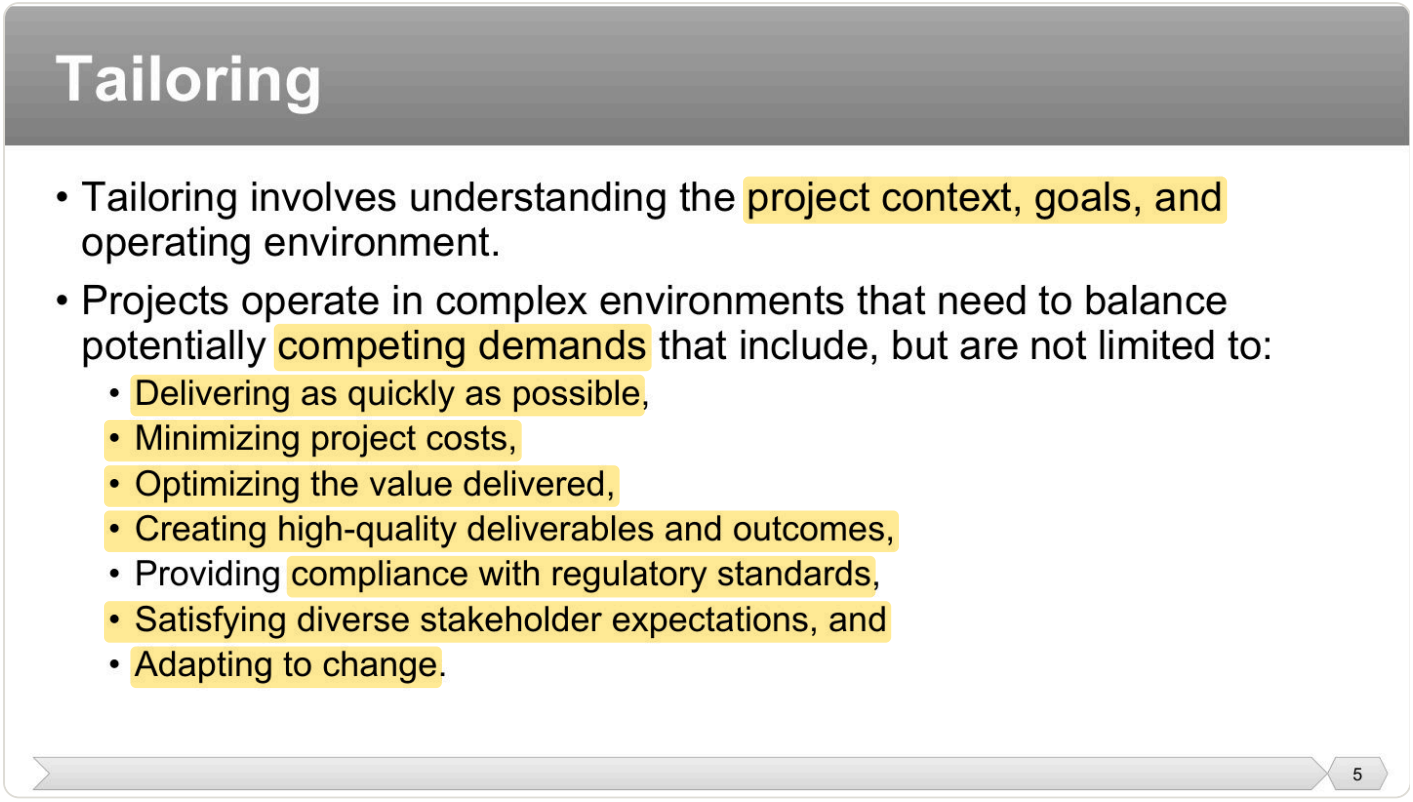
A Choosing the cheapest vendor for each task

B Removing all documentation to save time

C Following one standard methodology on every project

D **The deliberate adaptation of the project management approach, governance and processes to the environment and the work**

 Tailoring



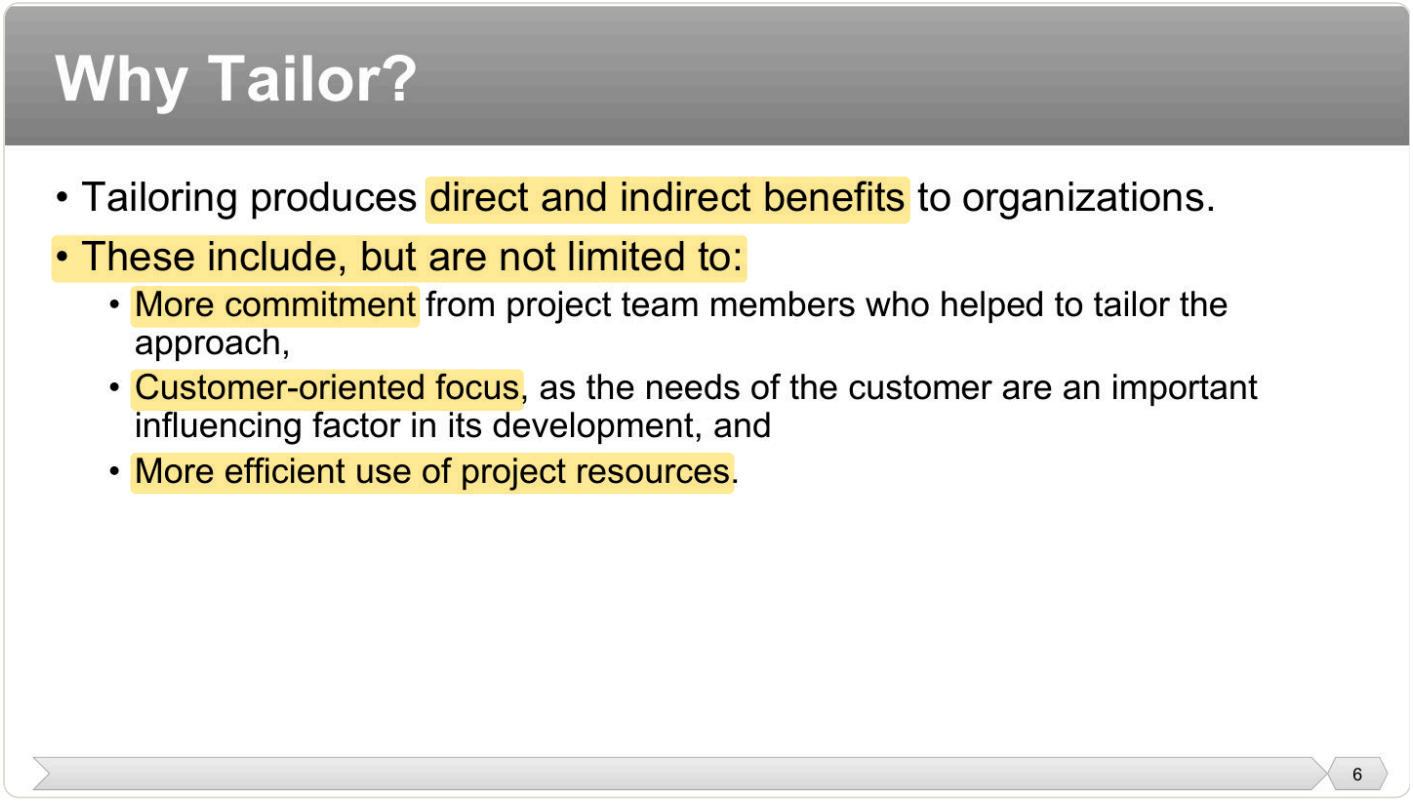
NOTES

1. Tailoring starts with understanding the project's **context, goals and operating environment**.
2. Projects must balance **competing demands**: speed, low cost, value, quality, regulatory compliance, diverse stakeholder expectations, and adapting to change. No project can maximize all seven at once (the fastest delivery is rarely the cheapest or the most thoroughly checked), which is why the approach has to be tailored to what matters most here.

TIPS & TRICKS

- **COMPADreS**: Creating high-quality deliverables, Optimizing value, Minimizing costs, Providing compliance, Adapting to change, Delivering quickly, Satisfying stakeholders.

 Why Tailor?



NOTES

1. Tailoring brings **direct and indirect benefits**:
2. **More commitment** from team members, because they helped shape the approach they now follow.
3. A **customer-oriented focus**, because the customer's needs drive the tailoring.
4. **More efficient use of resources**: effort goes where the project needs it, not into unneeded process.

EXAM ANGLE

Prediction Q1

Which is a stated benefit of tailoring?

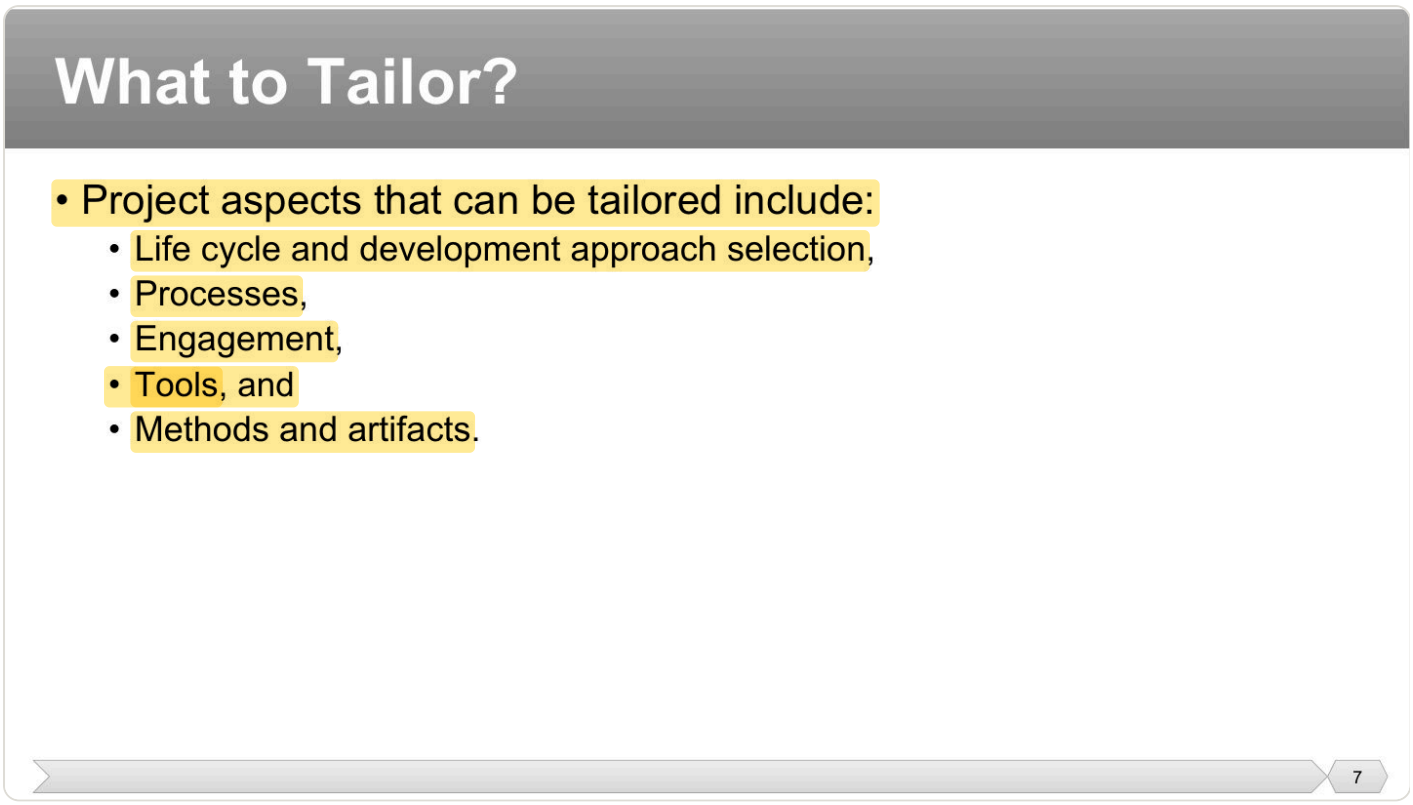
A **More commitment from team members who helped tailor the approach**

B A guaranteed fixed scope

C Eliminating the need for a project manager

D No further need for stakeholder input

 What to Tailor?



NOTES

1. Five aspects can be tailored (slides 8–12 take one each). **Life cycle and development approach**.
2. **Processes**.
3. **Engagement** of people.
4. **Tools**.
5. **Methods and artifacts**.

TIPS & TRICKS

- **TEMPLe**: Tools, Engagement, Methods and artifacts, Processes, Life cycle and development approach.

What to Tailor?

What to Tailor?

- Project aspects that can be tailored include:
  - Life cycle and development approach selection,
    - Deciding on a life cycle and the phases of the life cycle is an example of tailoring.
    - Additional tailoring can be done when selecting the development and delivery approach for the project.
    - For instance, building a new data center could involve
      - (a) the use of predictive approaches for the physical building construction and finishing and
      - (b) an iterative approach for understanding and establishing the computing capabilities required.
      - Viewed from a project level, this combination of approaches represents a hybrid approach, but the construction team and the computing team may only experience a predictive or iterative development approach.

8

NOTES

1. Choosing the **life cycle** and its phases is a form of tailoring.
2. So is choosing the **development and delivery approach**.
3. Example: a **new data center** uses a predictive approach for the building's construction and finishing, and an iterative approach to discover and establish the computing capabilities it needs.
4. At the project level that combination is **hybrid**, even though each team only experiences its own approach (the construction team sees predictive, the computing team sees iterative).

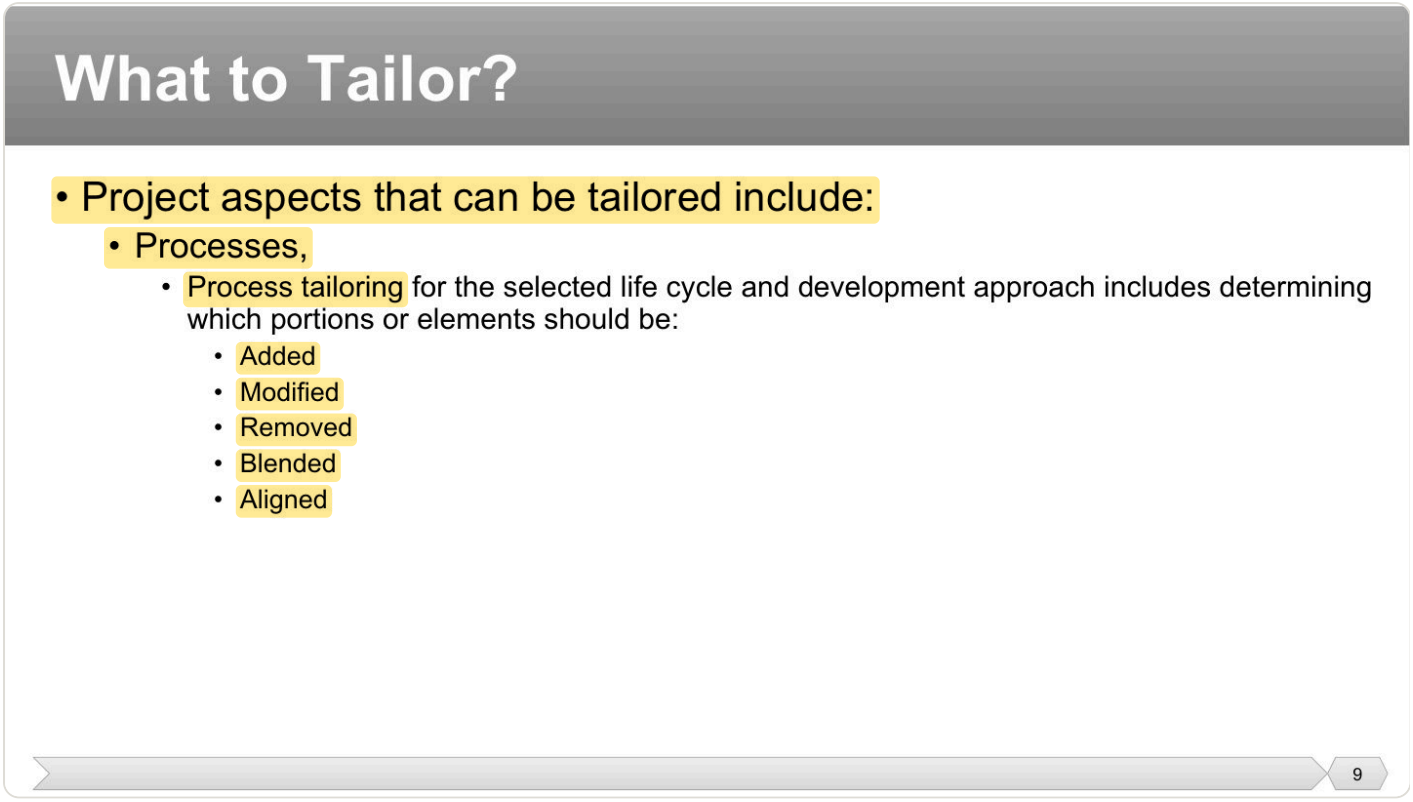
EXAM ANGLE

Prediction Q1

A new data center is built in fixed construction phases, while its computing capabilities are worked out in iterations. What is the overall approach?

- A **Hybrid**
- B Incremental only
- C Predictive
- D Adaptive

 What to Tailor?



NOTES

1. **Process tailoring** decides which parts of the chosen approach's processes are:
2. **Added** (extra rigor, e.g. a security review), **modified** (changed to fit, e.g. a lighter weekly report), **removed** (dropped as unnecessary), **blended** (elements from different methods combined into one, e.g. Scrum ceremonies with a predictive budget process) or **aligned** (harmonized so definitions and terms are consistent across teams).

TIPS & TRICKS

- **A RuMBA**: Aligned, Removed, Modified, Blended, Added.

EXAM ANGLE

Prediction Q1

Combining elements of two different methods into one process is which process-tailoring action?

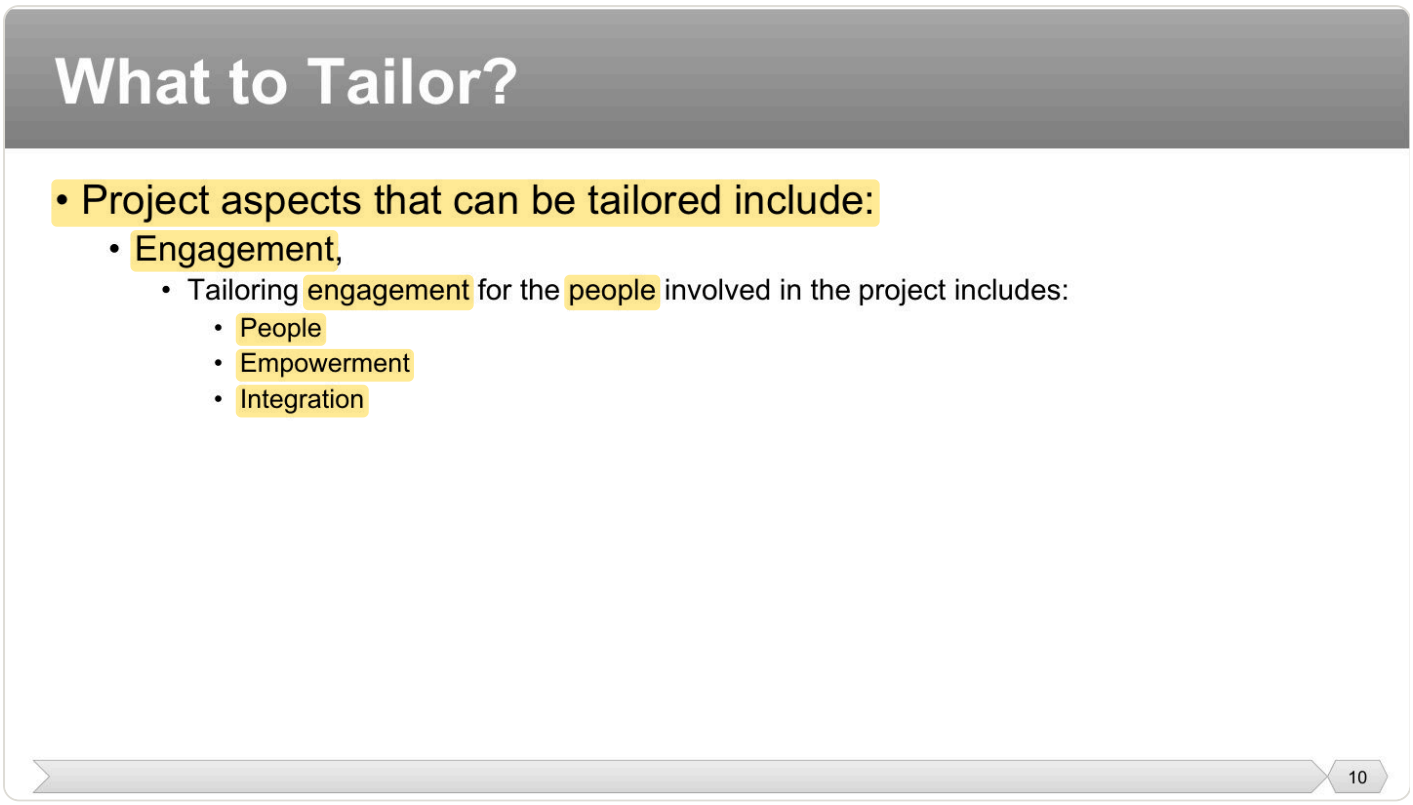
A Adding

B Aligning

C Removing

D **Blending**

 What to Tailor?



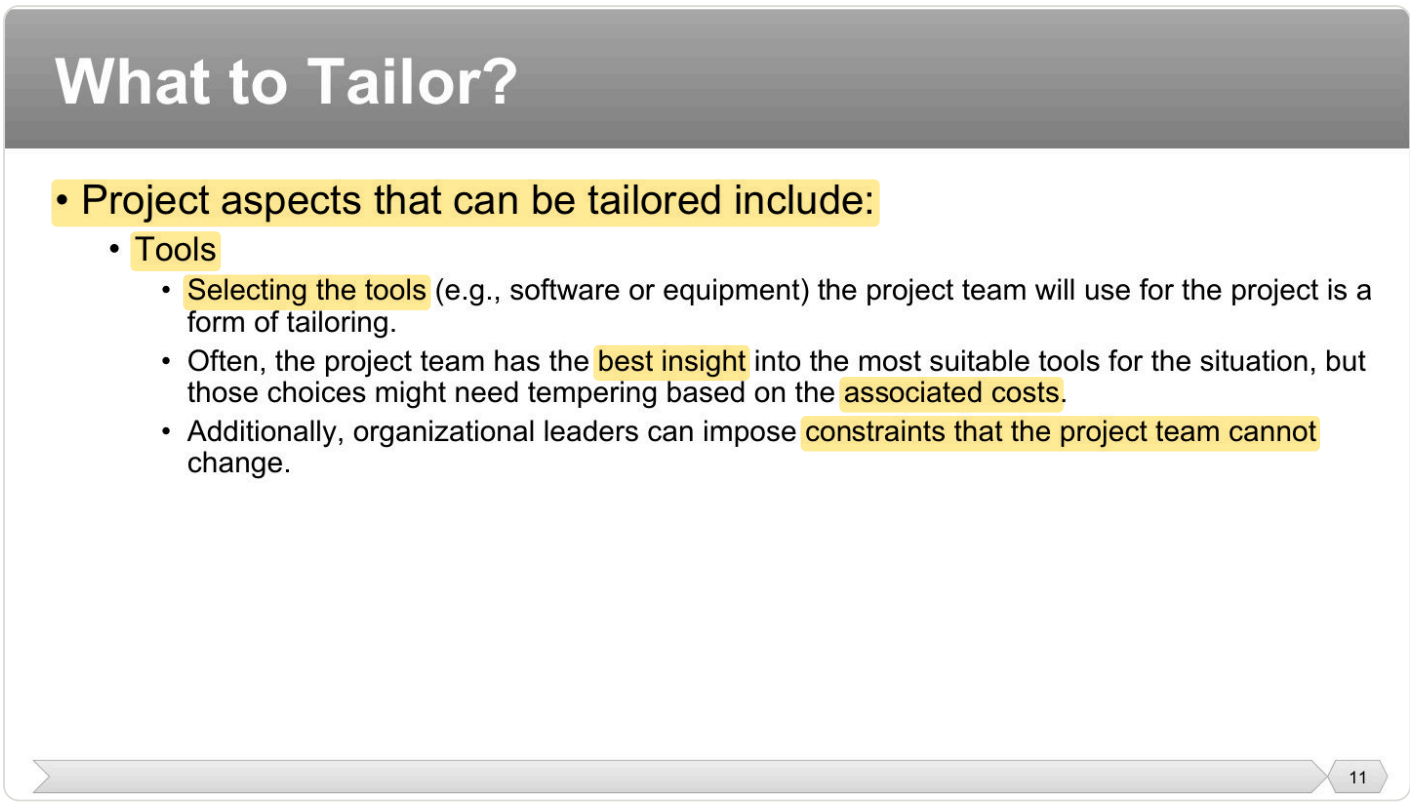
NOTES

1. **Engagement** tailoring decides how people are involved:
2. **People**: who is on the team and with which skills and capabilities. **Empowerment**: how much responsibility and local decision-making the team gets (a mature team gets more). **Integration**: how contractors, partners and other organizations are brought into the team.

TIPS & TRICKS

- **PIE**: People, Integration, Empowerment.

 What to Tailor?



NOTES

1. Choosing the **tools** (software, equipment) is tailoring.
2. The team usually has the **best insight** into which tools suit the work, but choices may be tempered by their **cost**.
3. Organizational leaders may also impose **constraints** the team can't change (a mandated corporate tool).

SLIDE 12 / 41 **What to Tailor?**

What to Tailor?

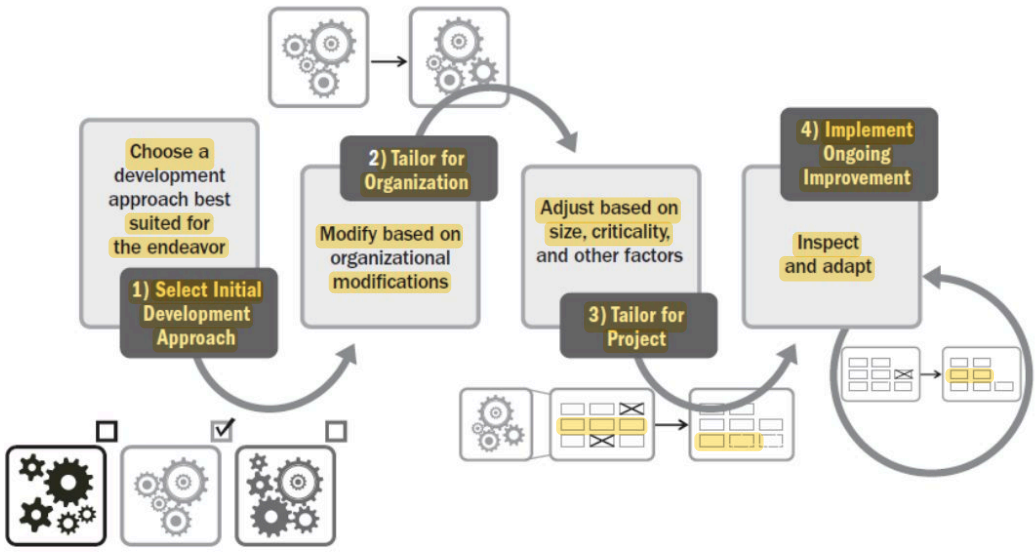
- Project aspects that can be tailored include:
  - Methods and artifacts
    - Tailoring the means that will be used to achieve the project outcomes is performed so that the methods are suited for the environment and the culture.
    - Tailoring the documents, templates, and other artifacts that will be used on the project helps to make sure the artifacts are appropriate for the project and the organization.

NOTES

1. **Methods** (the means of achieving outcomes) are tailored to fit the **environment and culture**.
2. **Artifacts** (documents, templates) are tailored so they are appropriate for the project and organization: a two-page charter for a small internal project, a full one for a regulated contract.

SLIDE 13 / 41 **The Tailoring Process**

The Tailoring Process



NOTES

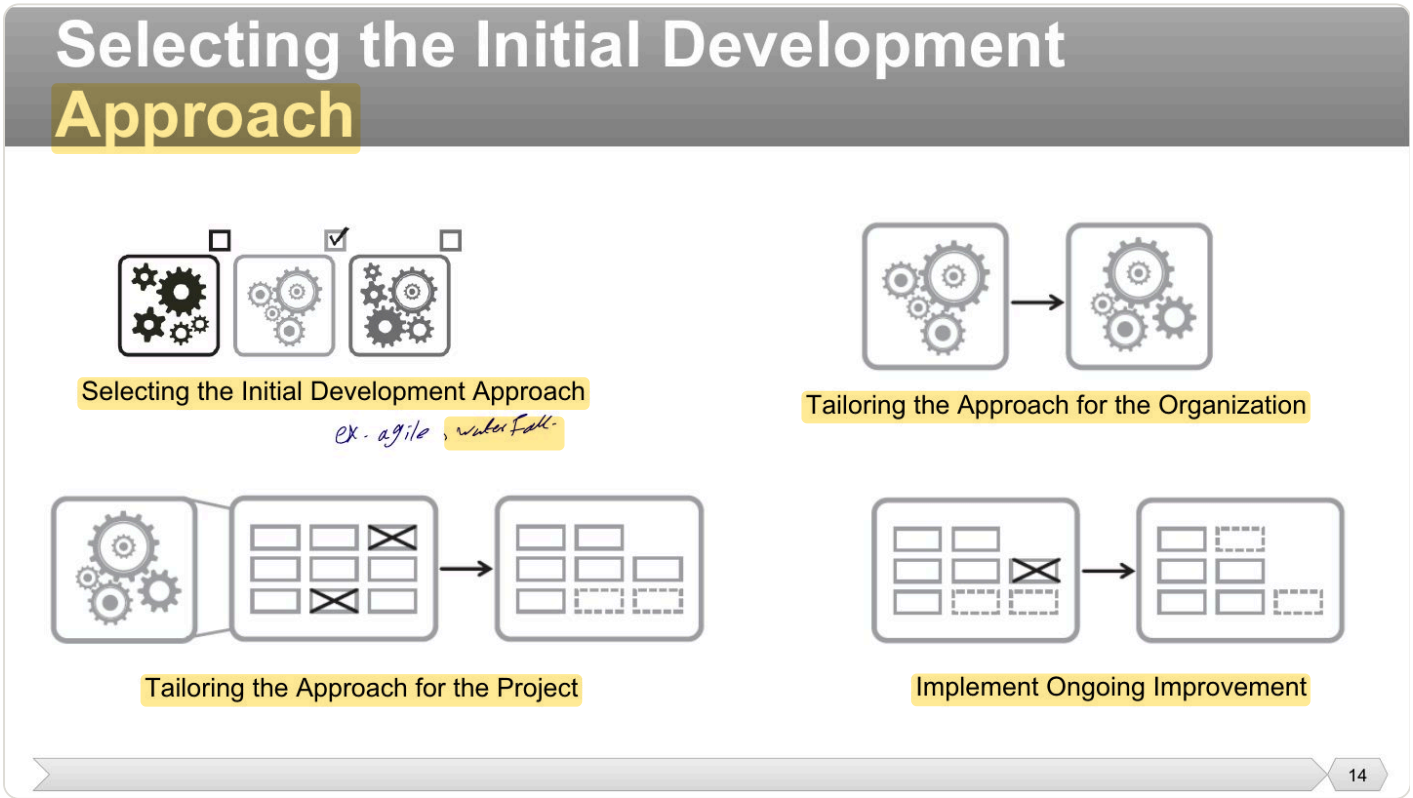
1. The tailoring process, step **1**: **select the initial development approach**: choose the approach best suited to the endeavor (the ticked set of gears).
2. Step **2**: **tailor for the organization**: modify it to fit organizational requirements and constraints.
3. Step **3**: **tailor for the project**: adjust for size, criticality and other project factors.
4. Step **4**: **implement ongoing improvement**: inspect and adapt; the loop arrow shows it never ends.

EXAM ANGLE

Prediction Q1

Put the four steps of the tailoring process in order.
Answer: (1) Select the initial development approach; (2) tailor the approach for the organization; (3) tailor it for the project; (4) implement ongoing improvement (inspect and adapt throughout the project).

SLIDE 14 / 41 **Selecting the Initial Development**

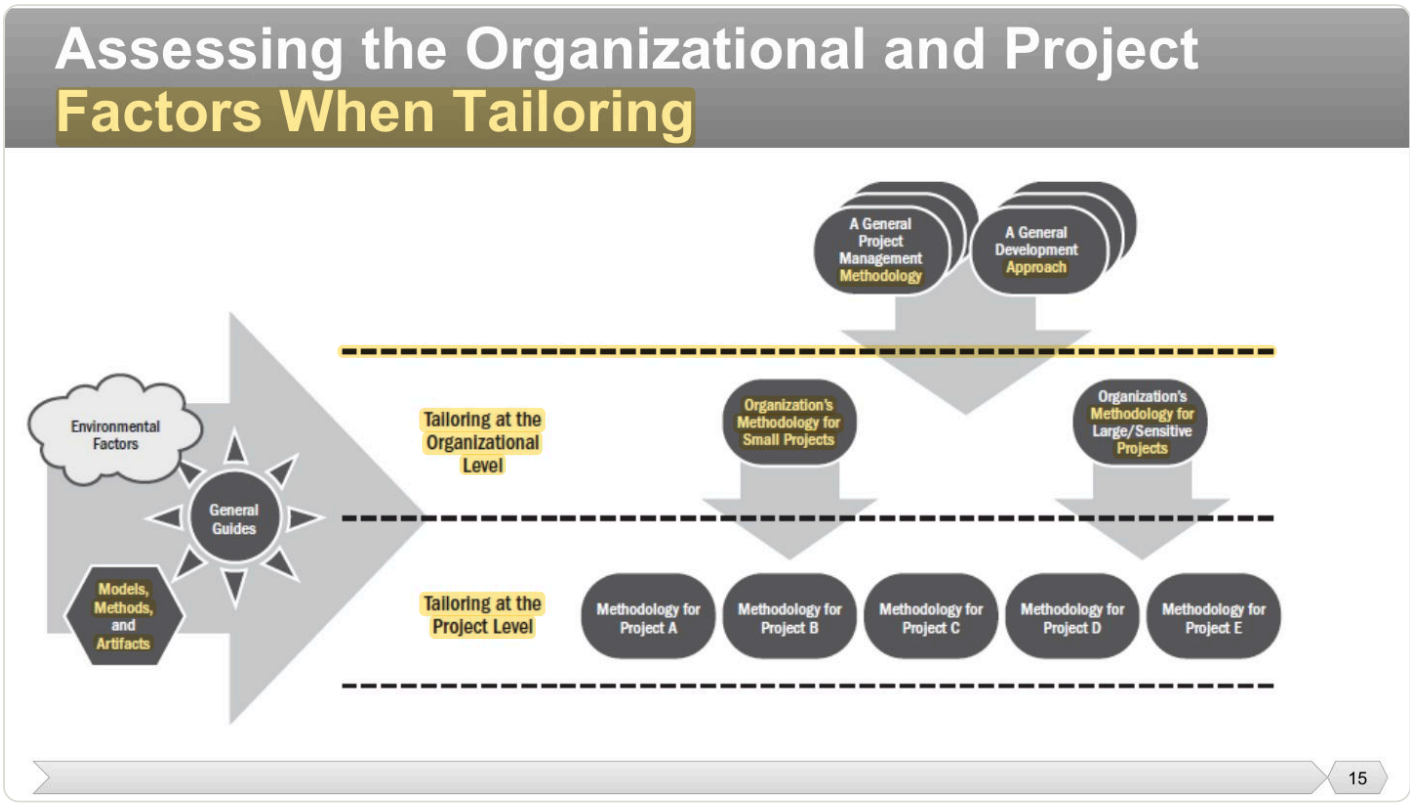


NOTES

1. Step 1 in pictures: choose the base approach, e.g. agile or waterfall (the handwritten note).
2. Step 2: one set of gears becomes the organization's adapted version.
3. Step 3: elements are removed (the crossed boxes) and adjusted for this project.

4 Step 4: the remaining process keeps being refined.

SLIDE 15 / 41 Assessing the Organizational and Project



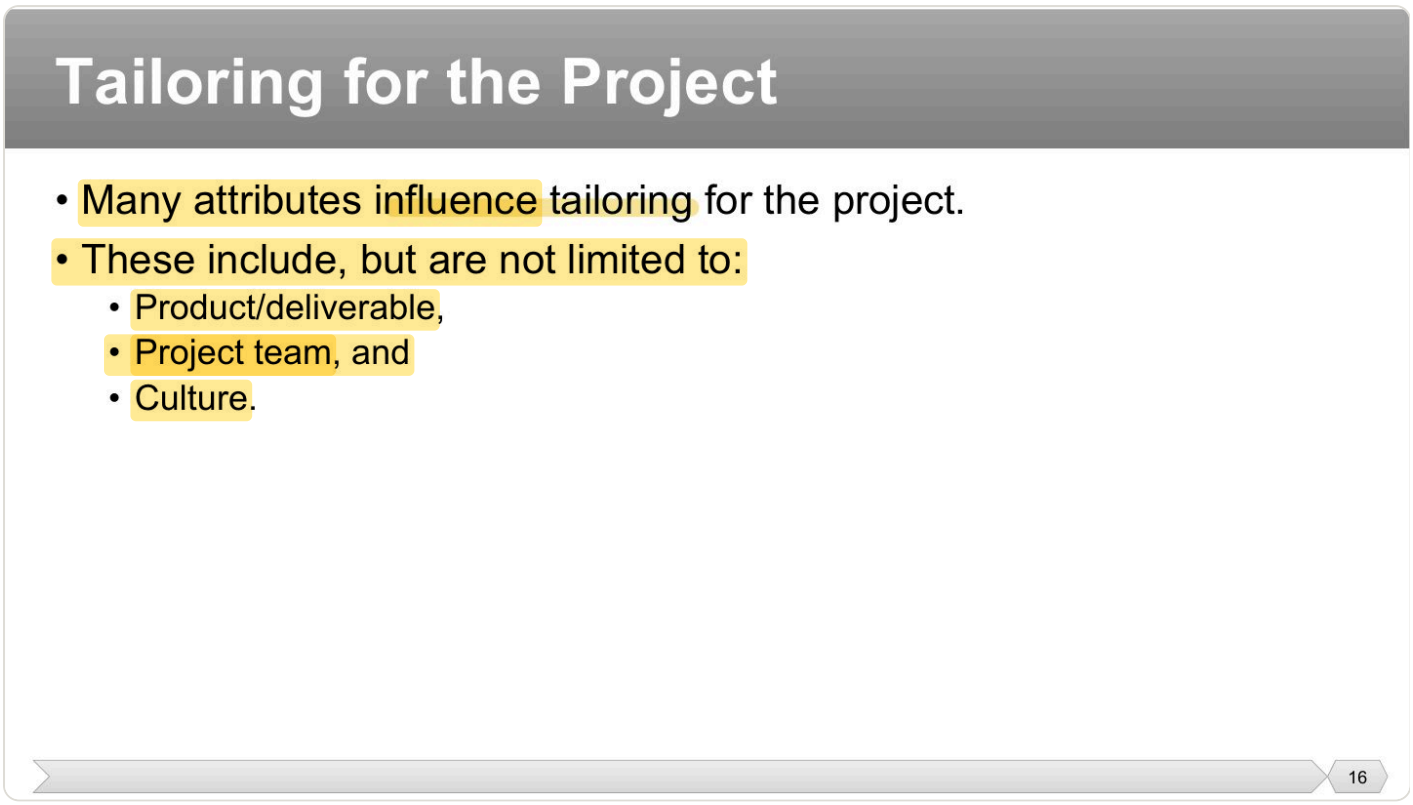
NOTES

1 Tailoring happens at **two levels**. At the **organizational** level, a general PM methodology and development approach are adapted into the organization's own methodologies, e.g. one for small projects and one for large or sensitive projects.

2 At the **project** level, each project tailors further (methodologies for projects A–E differ).

3 Both levels are guided by **environmental factors**, general guides, and the available models, methods and artifacts.

 Tailoring for the Project



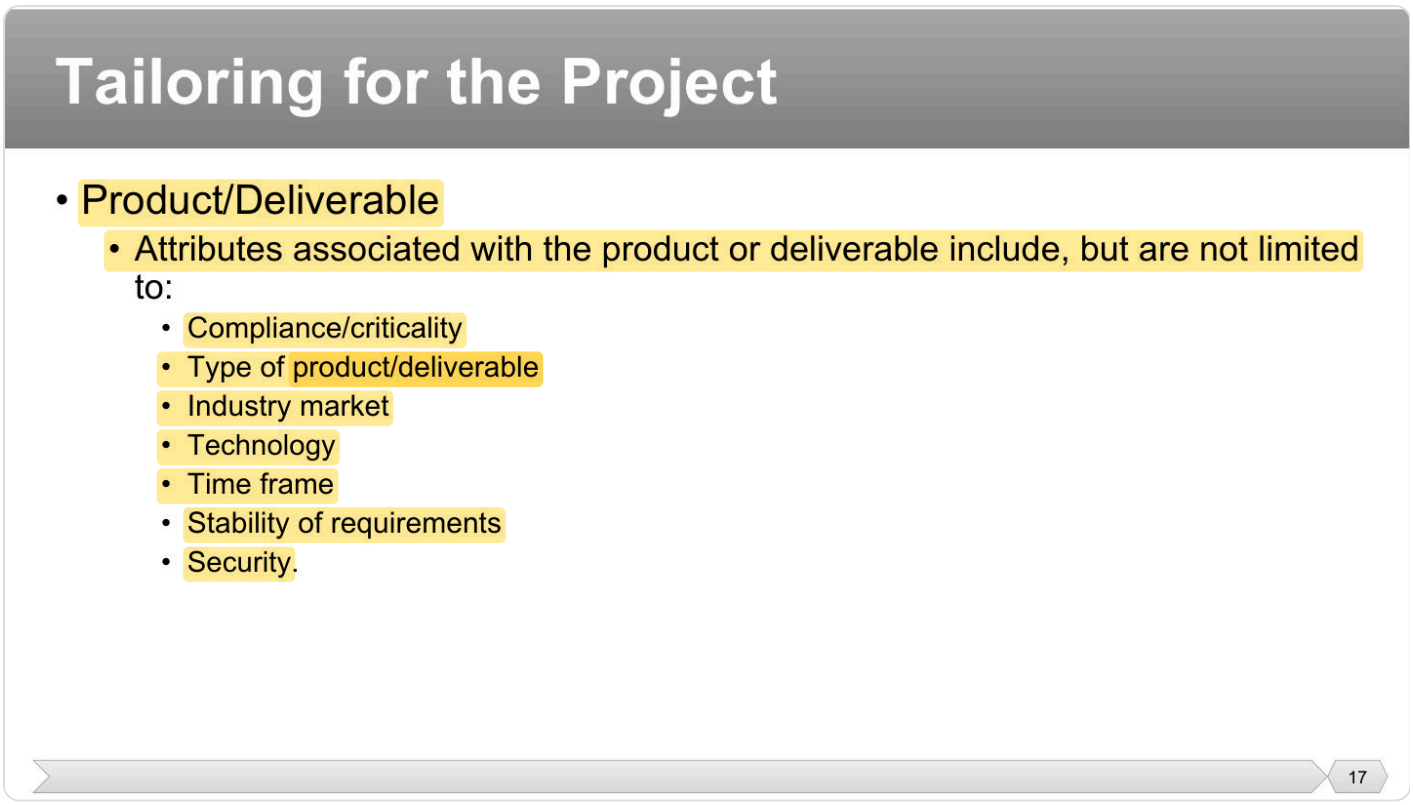
NOTES

1. Tailoring for the project depends on many attributes, grouped into three:
2. **Product/deliverable** (slide 17), **project team** (slide 18) and **culture** (slide 19).

TIPS & TRICKS

- **CoPPer**: Culture, Project team, Product or deliverable.

 Tailoring for the Project



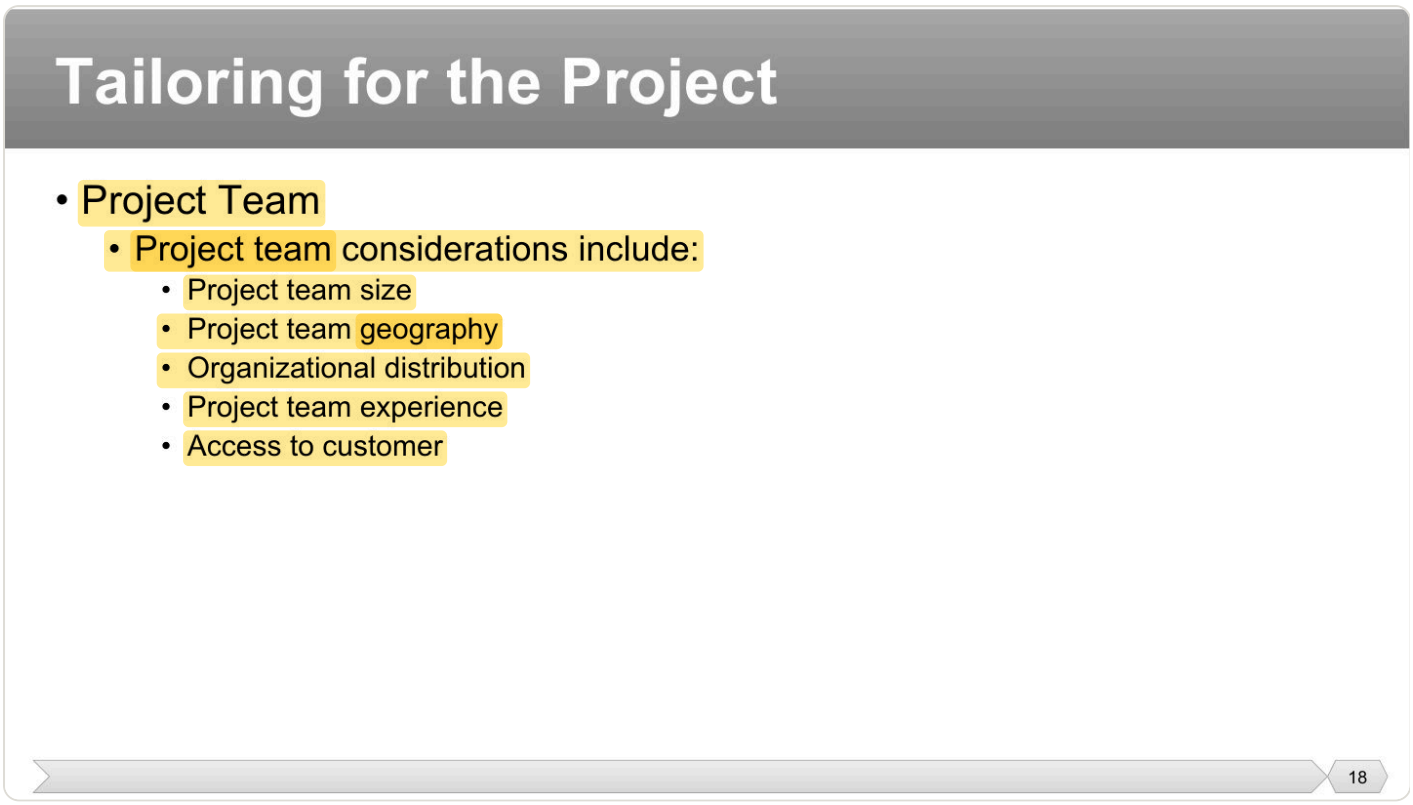
NOTES

1. **Product/deliverable** attributes.
2. **Compliance and criticality**: how much rigor and assurance are needed. Also the type of product, the industry market, the technology and the time frame.
3. **Stability of requirements** (stable favours predictive, volatile favours adaptive) and **security** needs.

TIPS & TRICKS

- Sentence, in slide order: **C**ats **T**ake **I**n **T**oo **T**iny **S**nacks, **S**adly. Compliance, Type of product, Industry market, Technology, Time frame, Stability of requirements, Security.

 Tailoring for the Project



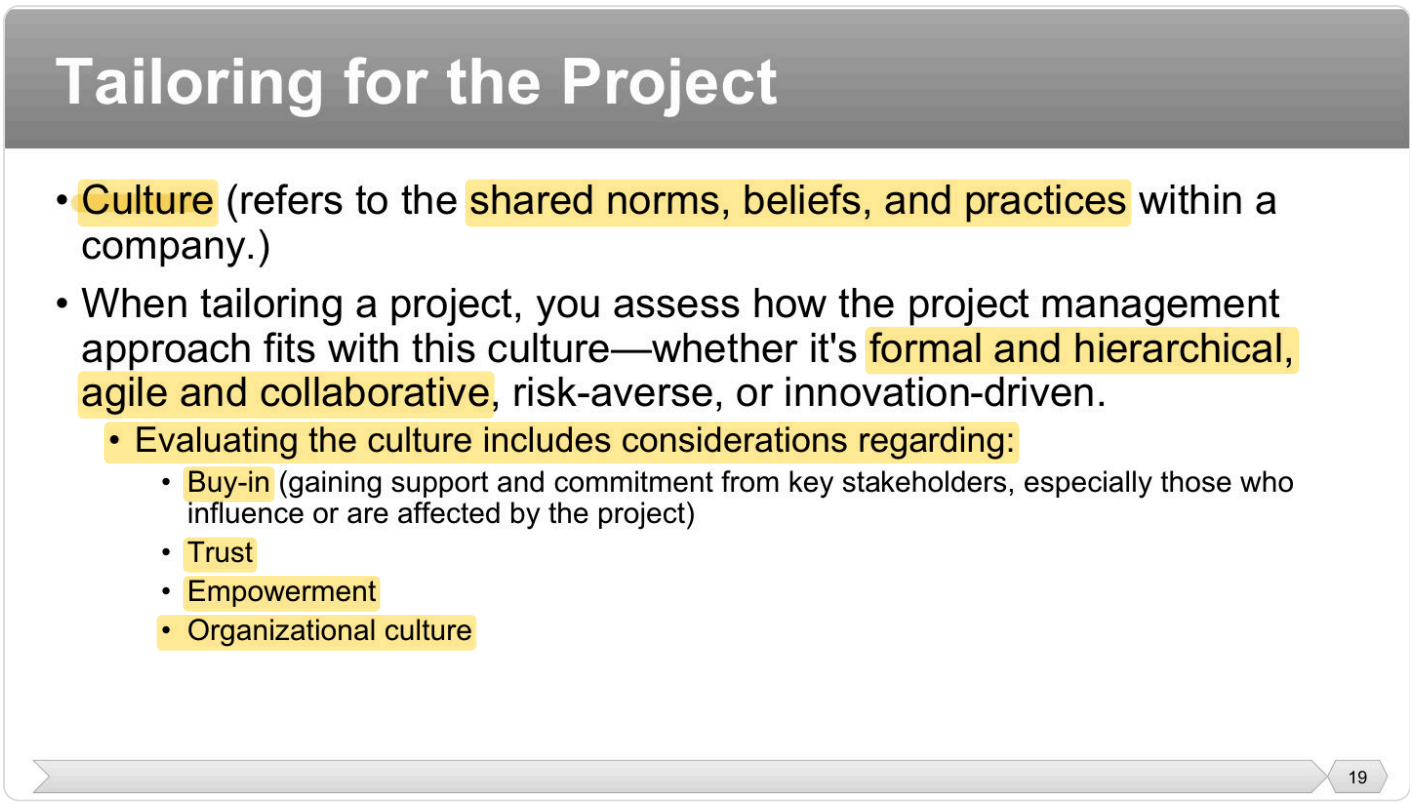
NOTES

1. **Project team** considerations.
2. Team **size** and **geography** (co-located or distributed), and how it is distributed across the organization.
3. Team **experience** (an experienced team needs less prescribed process) and **access to the customer** (adaptive approaches need frequent customer input).

TIPS & TRICKS

- **SAGE O**wl: Size of the team, Access to customer, Geography, Experience, Organizational distribution.

 Tailoring for the Project



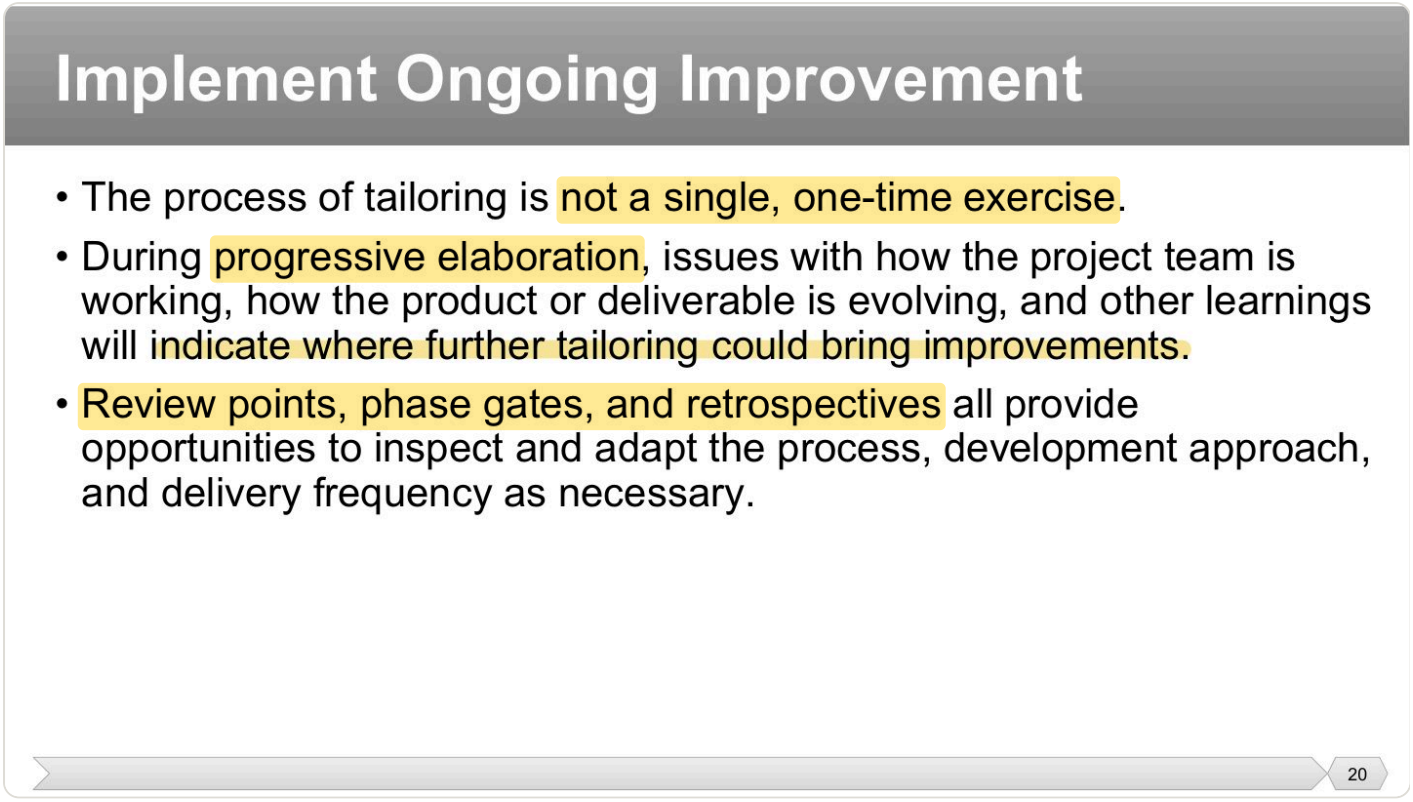
NOTES

1. **Culture** is the organization's **shared norms, beliefs and practices**.
2. Tailoring checks how the approach fits it: is the culture **formal and hierarchical, agile and collaborative**, risk-averse or innovation-driven? An agile process forced onto a strictly hierarchical culture struggles.
3. Evaluate **buy-in** (support from key stakeholders), **trust**, **empowerment** and the organizational culture as a whole.

TIPS & TRICKS

- **TO BE**: Trust, Organizational culture, Buy-in, Empowerment.

 Implement Ongoing Improvement



NOTES

1. Tailoring is **not a one-time exercise** done at the start.
2. During **progressive elaboration**, problems with how the team works or how the product evolves show where more tailoring would help.
3. **Review points, phase gates and retrospectives** are the opportunities to inspect and adapt the process, the approach and the delivery frequency.

EXAM ANGLE

Prediction Q1

True or false: tailoring is done once, when the project is initiated.

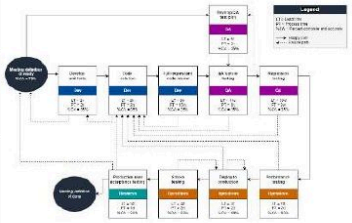
T True

F False

It continues throughout the project; reviews, phase gates and retrospectives are chances to adapt.

Situations and Tailoring Suggestions

Situation	Tailoring Suggestion
Poor quality deliverables	Add more feedback verification loops and quality assurance steps.
Team members unsure of how to proceed or undertake their work	Add more guidance, training, and verification steps.
Long delays waiting for approvals	Try streamlining approval decisions through fewer people authorized to make decisions up to certain value thresholds.
Too much work in progress or high rates of scrap	Use techniques like value stream mapping and kanban boards to visualize the work, identify the issues, and propose solutions.
Stakeholders are not engaged or share negative feedback	Evaluate whether sufficient information is being shared with stakeholders; feedback loops are present and working; and deeper engagement may work better than simply communicating.
Lack of visibility and understanding of project progress	Check to ensure appropriate measures are being collected, analyzed, shared, and discussed during team and stakeholder meetings; validate agreement with the measures within the team and with stakeholders.
Issues and/or risks for which the team is unprepared continue to surface, requiring the team to react rather than progress work	Explore root causes to identify whether there are related gaps in project processes or activities.



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NOTES

1. The pattern in this table: name the problem, then tailor in the missing control. **Poor quality deliverables** → add feedback, verification loops and quality-assurance steps. Members unsure how to proceed → add guidance, training and verification.
2. **Long approval delays** → streamline decisions: fewer people authorized to decide, up to value thresholds.
3. **Too much work in progress or scrap** → value-stream mapping and kanban boards to visualize the work and find the problems. Disengaged stakeholders → check information sharing and feedback loops; deeper engagement may work better than simply communicating.
4. **Lack of visibility of progress** → make sure the right measures are collected, analysed and discussed, and validated with the team and stakeholders. Recurring surprises → root-cause analysis of gaps in the process.

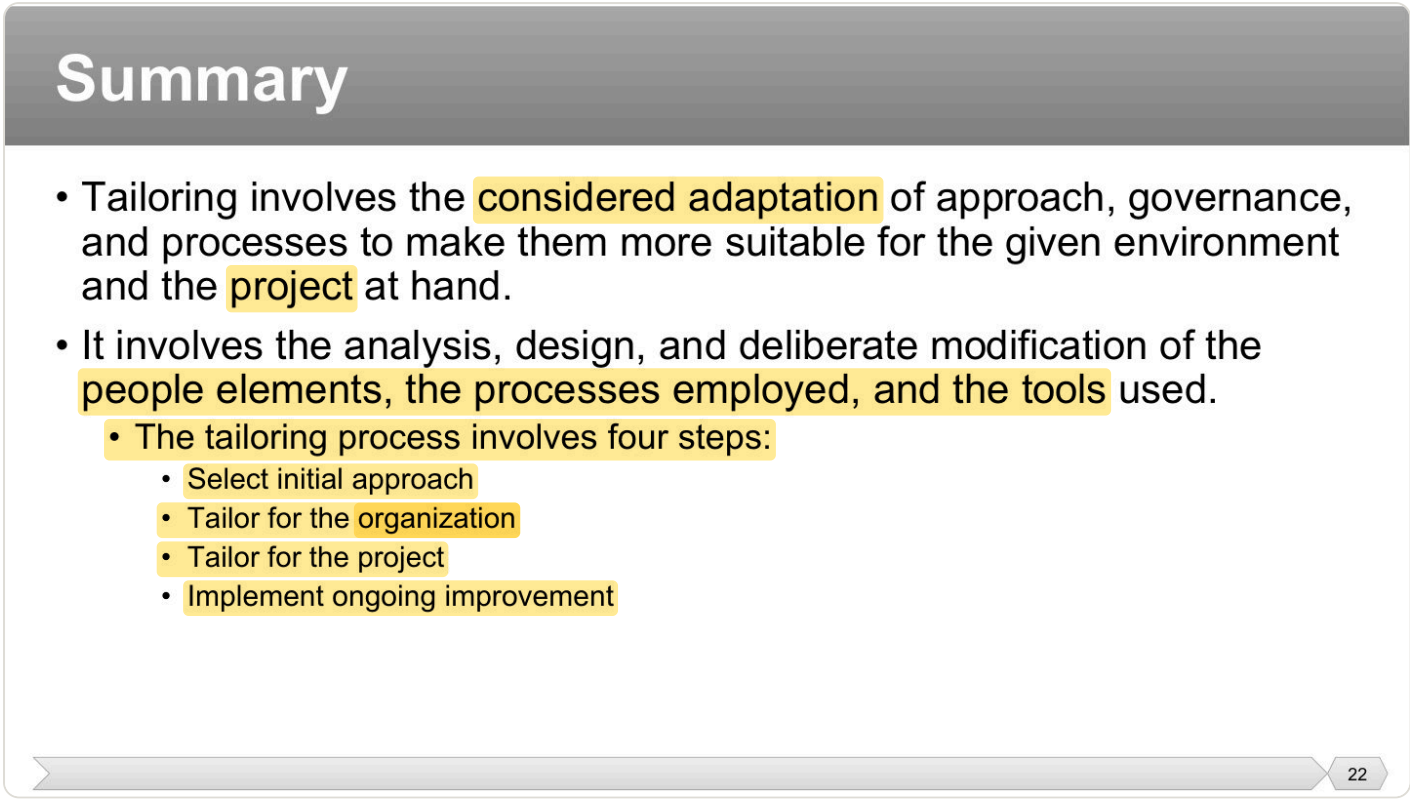
EXAM ANGLE

Prediction Q1

A team keeps waiting several days for approvals before it can proceed. What tailoring change does the table suggest?

Answer: Streamline approval decisions: authorize fewer people to make decisions, up to certain value thresholds, so routine approvals don't queue.

 Summary



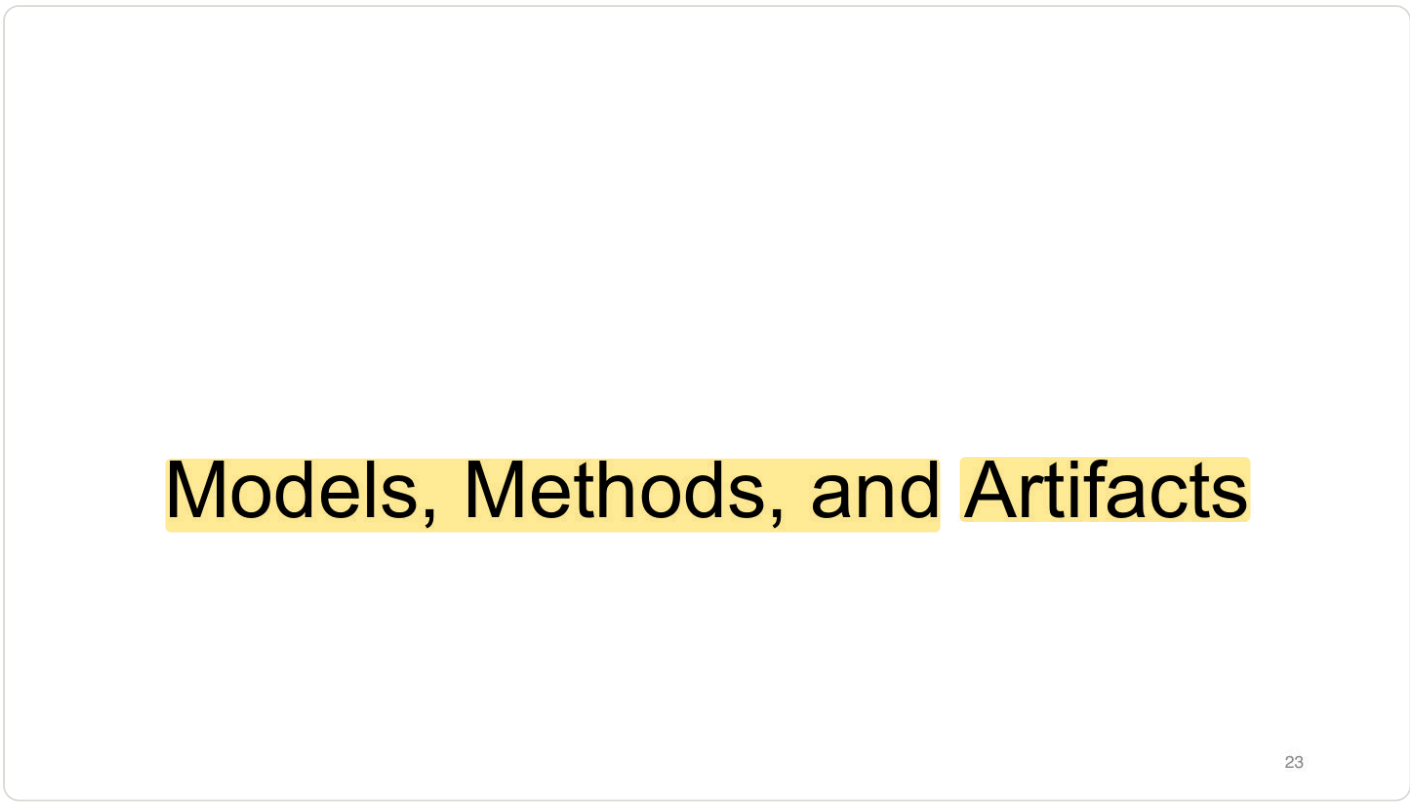
NOTES

1. Summary: tailoring is the **considered adaptation** of approach, governance and processes to the environment and project.
2. It involves analysing, designing and deliberately modifying the **people, processes and tools**.
3. Four steps: **select** the initial approach → tailor for the **organization** → tailor for the **project** → **implement ongoing improvement**.

TIPS & TRICKS

- The order matters here: **S**mart **T**eams **T**ailor **I**teratively. Select the initial approach, Tailor for the organization, Tailor for the project, Implement ongoing improvement.

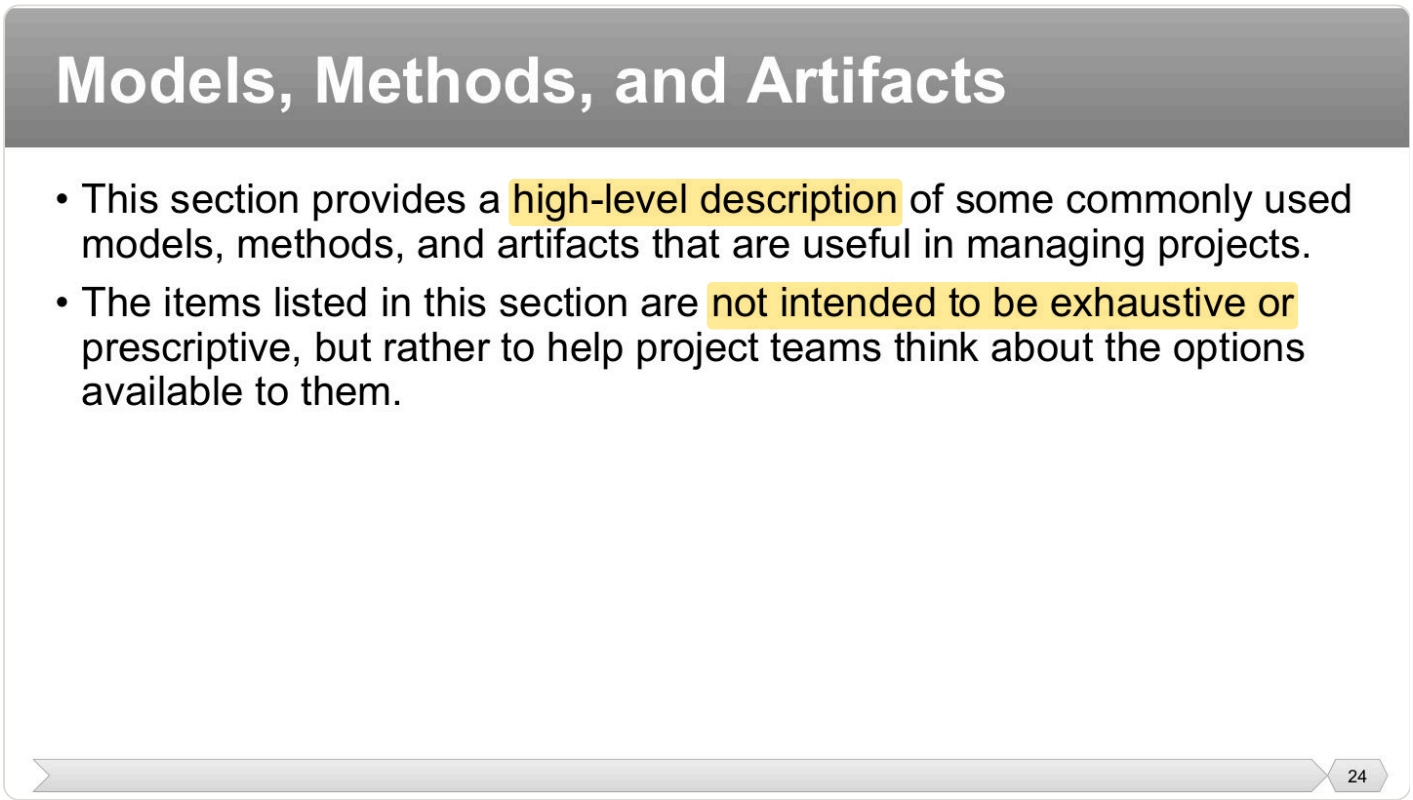
SLIDE 23 / 41 Models, Methods, and Artifacts (section)



NOTES

1. Section divider: models, methods and artifacts.

SLIDE 24 / 41 Models, Methods, and Artifacts

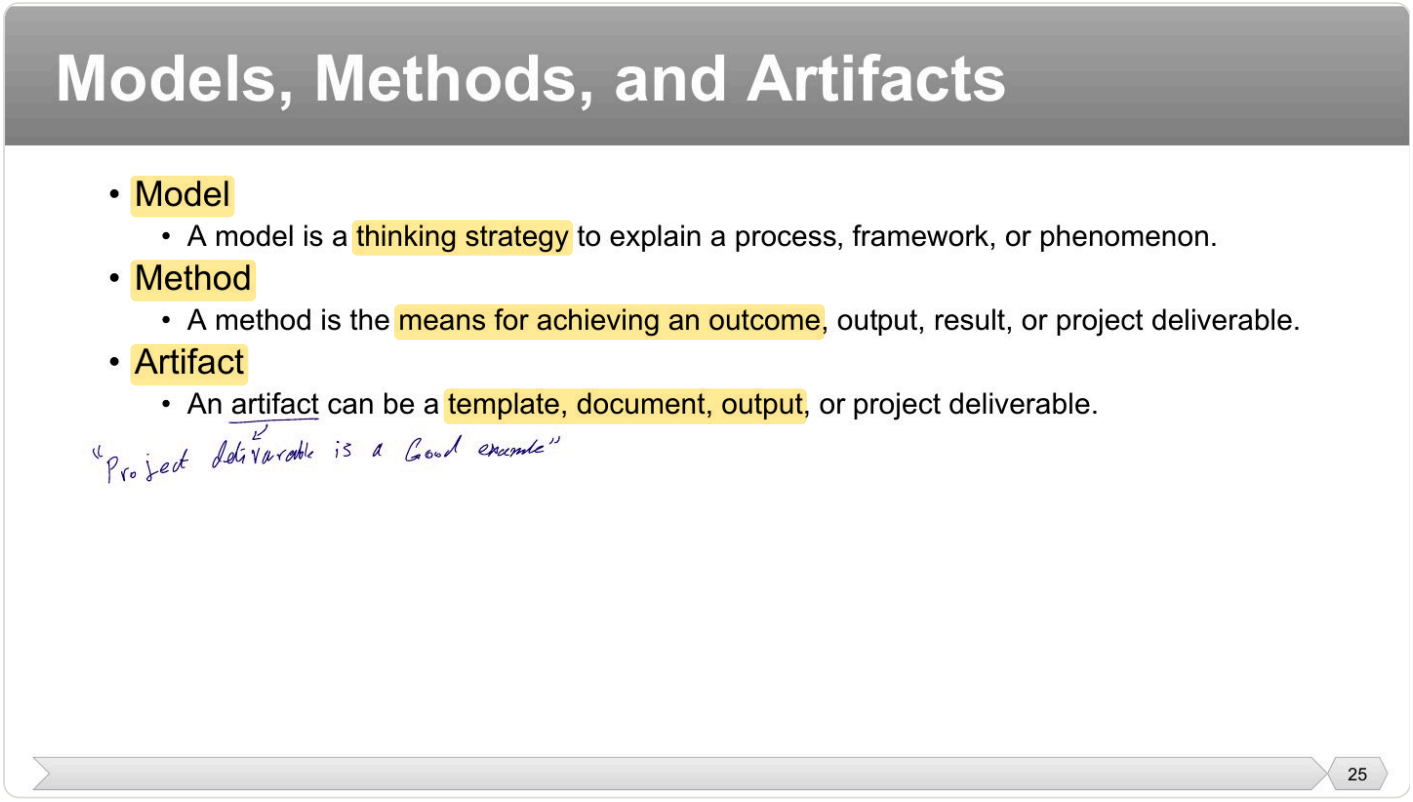


NOTES

1. This section gives a **high-level description** of common models, methods and artifacts.

2. It is **not exhaustive or prescriptive**: a menu of options to tailor from, not a checklist to use every time.

SLIDE 25 / 41 Models, Methods, and Artifacts



NOTES

1. A **model** is a **thinking strategy** that explains a process, framework or phenomenon (e.g. Tuckman explains how teams develop).
2. A **method** is the **means for achieving** an outcome, output, result or deliverable (e.g. estimating, a retrospective).
3. An **artifact** is a **template, document, output or deliverable** (e.g. a risk register, a project charter). Model = think, method = do, artifact = thing.

TIPS & TRICKS

- **MaMA**: Model, Method, Artifact.

EXAM ANGLE

Prediction

Q1

A risk register is best classified as a:

A Model

B Method

C Principle

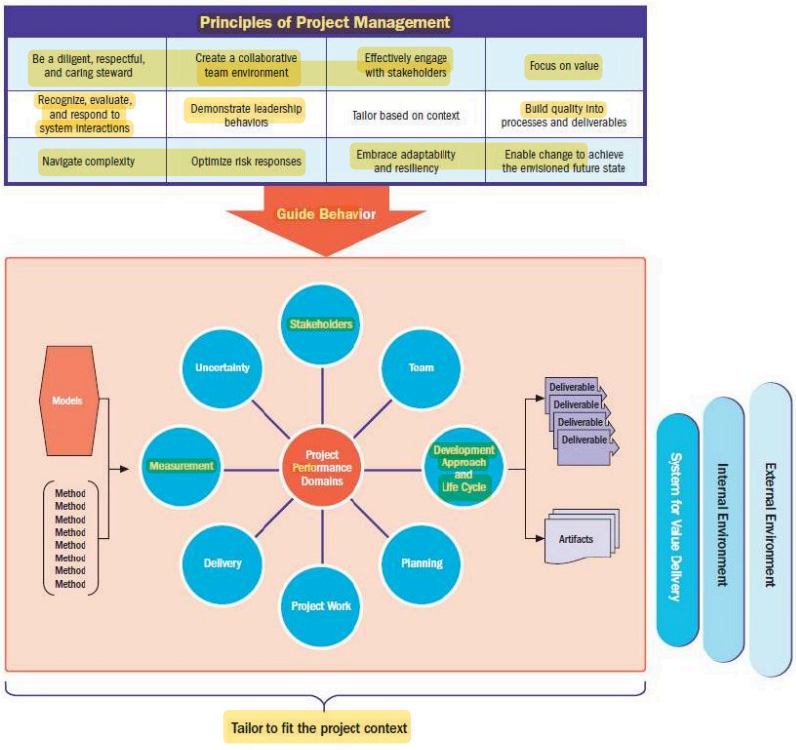
D **Artifact**

Past exam · Final 251 · Q1.4

Q2

Give an example of a project management tool or technique that you might use to manage a complex software development project, and explain how it could help you ensure project success.
Answer: A Work Breakdown Structure: it decomposes the complex system into smaller deliverables and work packages that can be estimated, assigned and tracked. Nothing is forgotten, ownership is clear, and the WBS is the base for the schedule, budget and risk analysis, so problems show up early and progress is measurable. (Other valid answers: Gantt chart, CPM, risk register, Scrum board with a burndown chart, each with its benefit explained.)

SLIDE 26 / 41 Principles, domains and the toolbox together

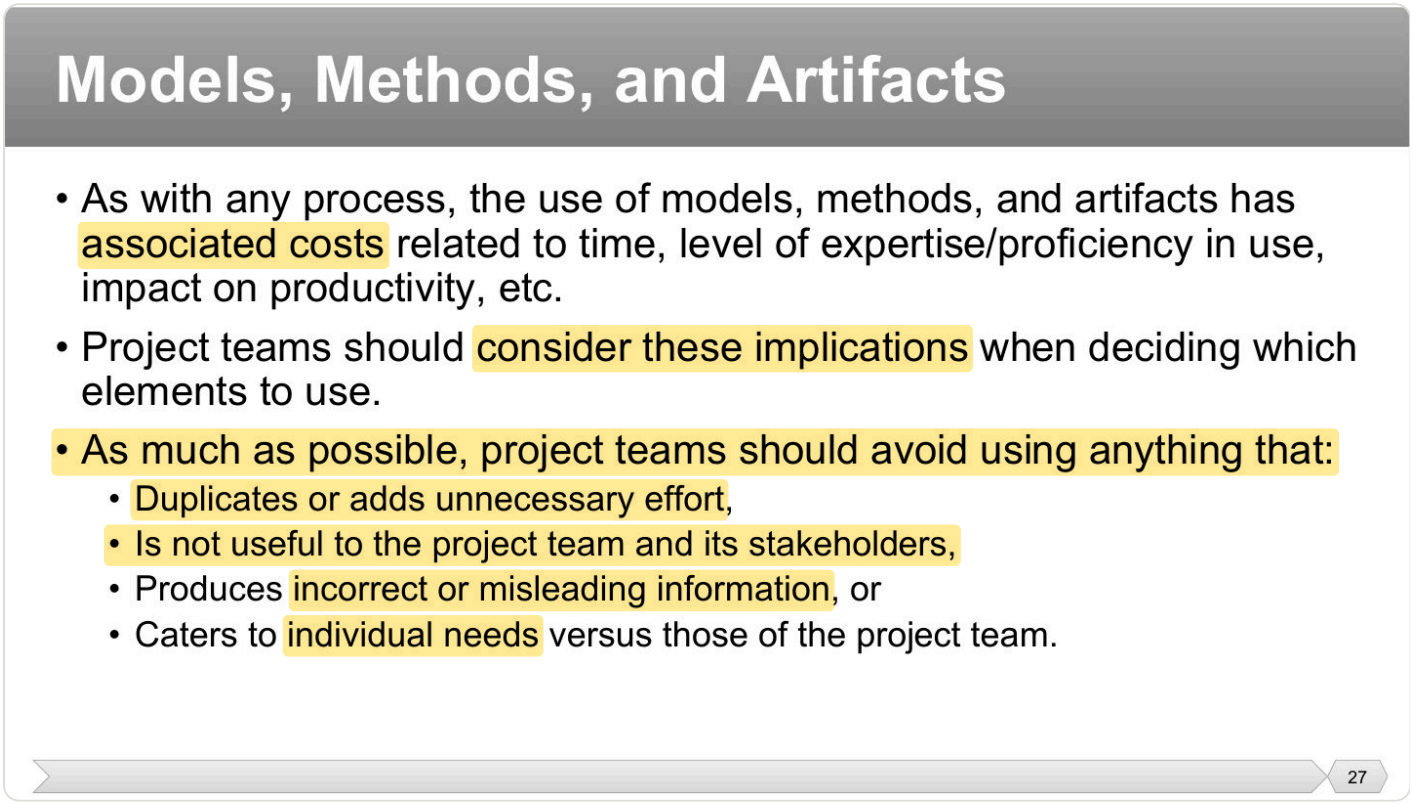


26

NOTES

1. The whole PMBOK 7 picture in one diagram. At the top, the **12 principles**...
2. ...**guide behavior** in...
3. ...the **8 performance domains** (the circle), which are carried out with **models, methods and artifacts** (left and right) to produce deliverables. Everything sits inside the system for value delivery and the internal and external environments, and is **tailored to fit the project context** (bottom).

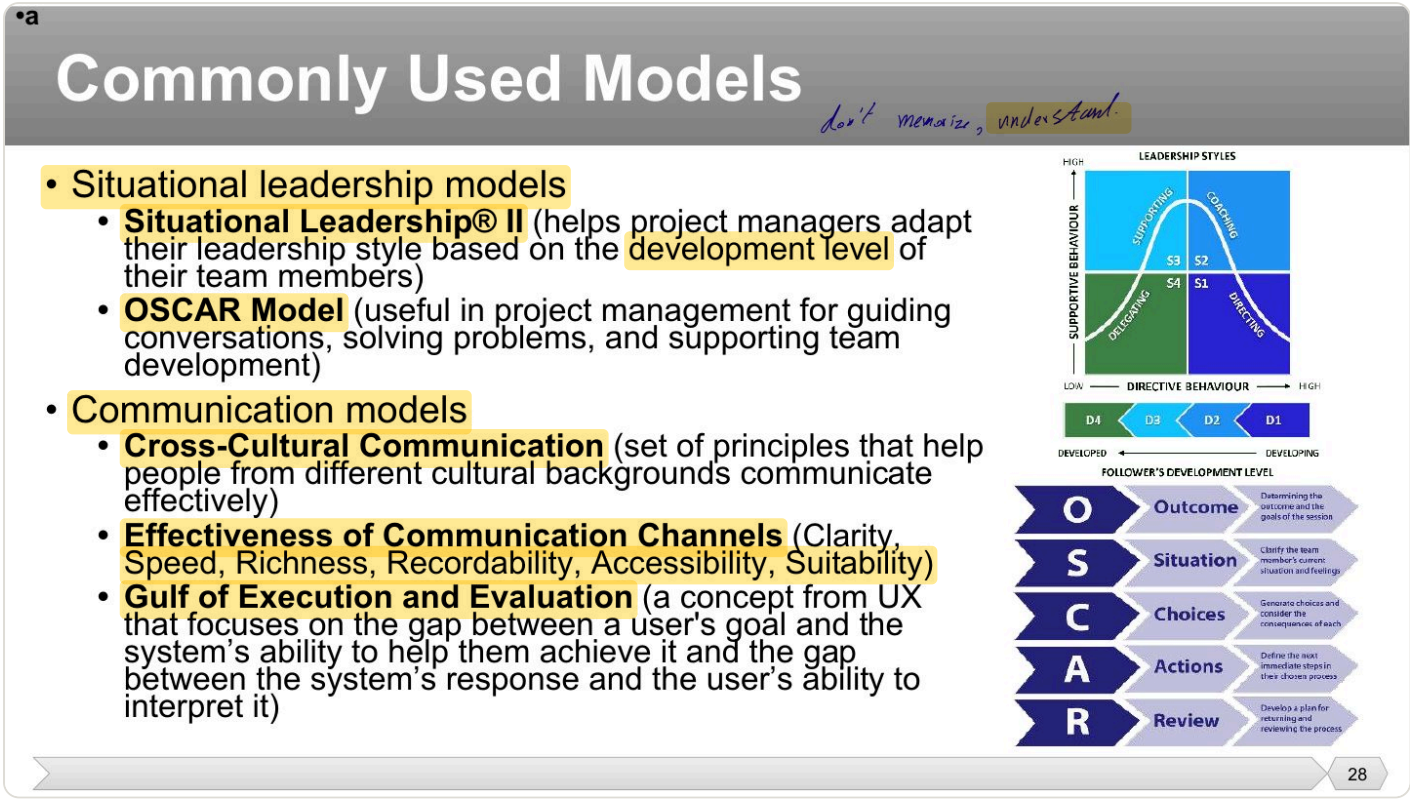
SLIDE 27 / 41 Models, Methods, and Artifacts



NOTES

1. Every model, method and artifact has **costs**: time, the expertise needed to use it, the impact on productivity.
2. Teams should **weigh these** before adopting one.
3. Avoid anything that **duplicates** work or adds unnecessary effort, isn't useful to the team and stakeholders, produces **incorrect or misleading** information, or serves one **individual's** needs rather than the team's (a report one manager likes but nobody reads).

SLIDE 28 / 41 Commonly Used Models



NOTES

1. **Situational Leadership II**: the PM adapts style to each member's **development level** (competence and commitment): directing a beginner, coaching, supporting, then delegating to an expert. The instructor's note: understand these models, don't memorize them.

2. **OSCAR** (Outcome, Situation, Choices/consequences, Actions, Review): a coaching model for guiding conversations, solving problems and developing people.

3. **Communication models**: cross-cultural communication (principles for communicating across cultures) and the effectiveness of communication **channels** (clarity, speed, richness, recordability, accessibility, suitability): pick the channel that fits the message.

4. **Gulf of execution and evaluation** (from UX): the gap between a user's goal and what the system lets them do, and between the system's response and the user's ability to interpret it.

EXAM ANGLE

Prediction Q1

Which model helps a project manager adapt leadership style to each team member's development level?

A ADKAR

B Cynefin framework

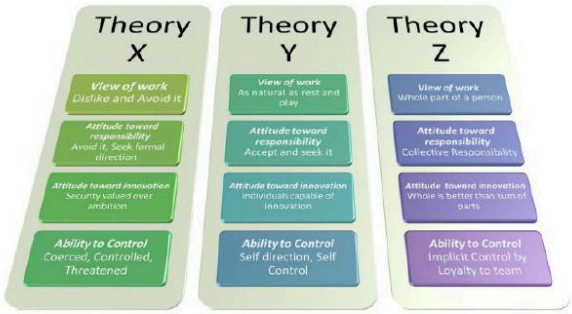
C **Situational Leadership II**

D Salience model

SLIDE 29 / 41 **Commonly Used Models**

Commonly Used Models

- Motivational models
  - Hygiene and Motivational Factors (Hygene: salary, benefits, working conditions, company policies, job security, etc. Motivational: Achievement, recognition, advancement, responsibility, the work itself, personal growth)
  - Intrinsic (to the work itself) versus Extrinsic Motivation ex · the team excites You ·
  - Theory of the hierarchy of Needs (physiological->safety->love/belonging->esteem->self-actualization)
  - Theory X, Theory Y, and Theory Z (These psychology of management styles models describe different assumptions about employee motivation and behavior)



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NOTES

1. **Herzberg's hygiene and motivational factors**: **hygiene** factors (salary, benefits, conditions, policies, job security) only prevent dissatisfaction; real **motivators** are achievement, recognition, advancement, responsibility, the work itself and growth. A pay rise is hygiene, not a motivator.
2. **Intrinsic** motivation comes from the work itself (it excites you); **extrinsic** from outside rewards (pay, grades).
3. **Maslow's hierarchy of needs**: physiological → safety → love/belonging → esteem → self-actualization; lower needs come first.

4 **Theory X**: people dislike work and must be controlled. **Theory Y**: people want to work and should be empowered. **Theory Z**: long-term employment, loyalty and involvement.

EXAM ANGLE

Prediction Q1

In Herzberg's model, which of these is a motivator rather than a hygiene factor?

A Job security

B Working conditions

C **Recognition**

D Salary

Prediction Q2

A manager assumes employees dislike work and must be closely supervised. Which theory does this reflect?

A **Theory X**

B Theory Y

C Theory Z

D Situational Leadership

 Commonly Used Models

Commonly Used Models

h

- Change models
  - ADKAR® Model (focuses on **individual transformation**)
  - The 8-Step Process for Leading Change (Create a Sense of Urgency -> Build a Guiding Coalition -> Form a Strategic Vision and Initiatives -> Enlist a Volunteer Army -> Enable Action by Removing Barriers -> Generate Short-Term Wins -> Sustain Acceleration -> Institute Change)
  - Virginia Satir Change Model (is a psychologically grounded framework that explains how individuals and organizations emotionally respond to change.)
  - Transition Model (a powerful framework for understanding the **emotional and psychological journey** people go through during change)



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NOTES

1. **ADKAR** focuses on **individual** change: Awareness → Desire → Knowledge → Ability → Reinforcement.
2. **Kotter's 8 steps** lead **organizational** change: create urgency → build a guiding coalition → form a vision → enlist volunteers → remove barriers → generate short-term wins → sustain acceleration → institute the change.
3. **Virginia Satir** model: how individuals and organizations **emotionally** respond to change (an initial dip in performance before improvement).
4. **Transition model** (Bridges): the emotional journey through change: ending, neutral zone, new beginning.

EXAM ANGLE

Prediction Q1

Which change model focuses on the individual, through awareness, desire, knowledge, ability and reinforcement?

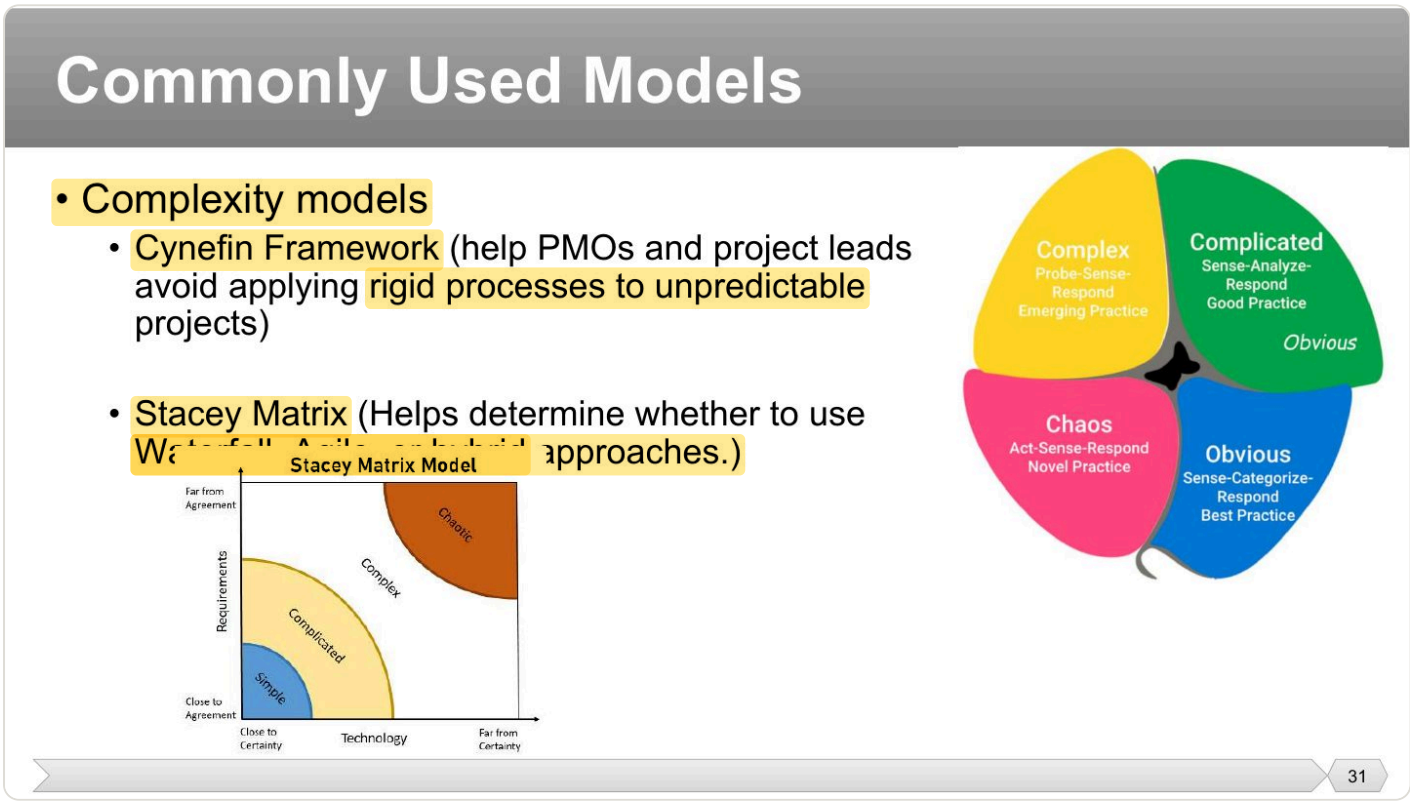
A Tuckman ladder

B **ADKAR**

C Stacey matrix

D Kotter's 8-step process

SLIDE 31 / 41 **Commonly Used Models**



NOTES

1. **Cynefin** sorts situations into obvious (apply best practice), complicated (analyse with experts), complex (probe and experiment) and chaotic (act first). It stops teams applying rigid processes to unpredictable projects.

2. The **Stacey matrix** places a project by how uncertain its requirements and its technology are: the more uncertain both are, the more an agile or adaptive approach fits. It helps choose waterfall, agile or hybrid.

EXAM ANGLE

Prediction Q1

Which model helps decide between waterfall, agile and hybrid approaches based on how uncertain the requirements and technology are?

A OSCAR

B ADKAR

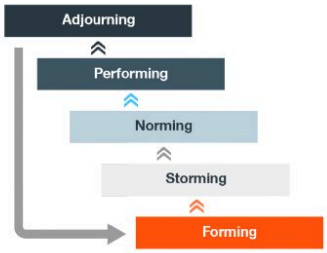
C Salience model

D **Stacey matrix**

SLIDE 32 / 41 **Commonly Used Models**

Commonly Used Models

- Project team development models
  - Tuckman Ladder (a classic model for understanding how project teams evolve over time.)



  - Drexler/Sibbet Team Performance Model (is a comprehensive framework that maps out the **seven predictable stages** teams go through: orientation -> trust building -> goal clarification -> commitment -> implementation -> high performance -> renewal)

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NOTES

1. **Tuckman ladder**: forming → storming → norming → performing → adjourning. Chapter 3 covers each stage, and the midterm asked which stage has members competing for leadership (storming).
2. **Drexler/Sibbet**: seven stages: orientation → trust building → goal clarification → commitment → implementation → high performance → renewal.

EXAM ANGLE

Past exam · Midterm 251 · V1 · MCQ 7 Q1

According to Tuckman's model of team development, which stage is characterized by members competing for leadership and struggling with group processes?

A Storming

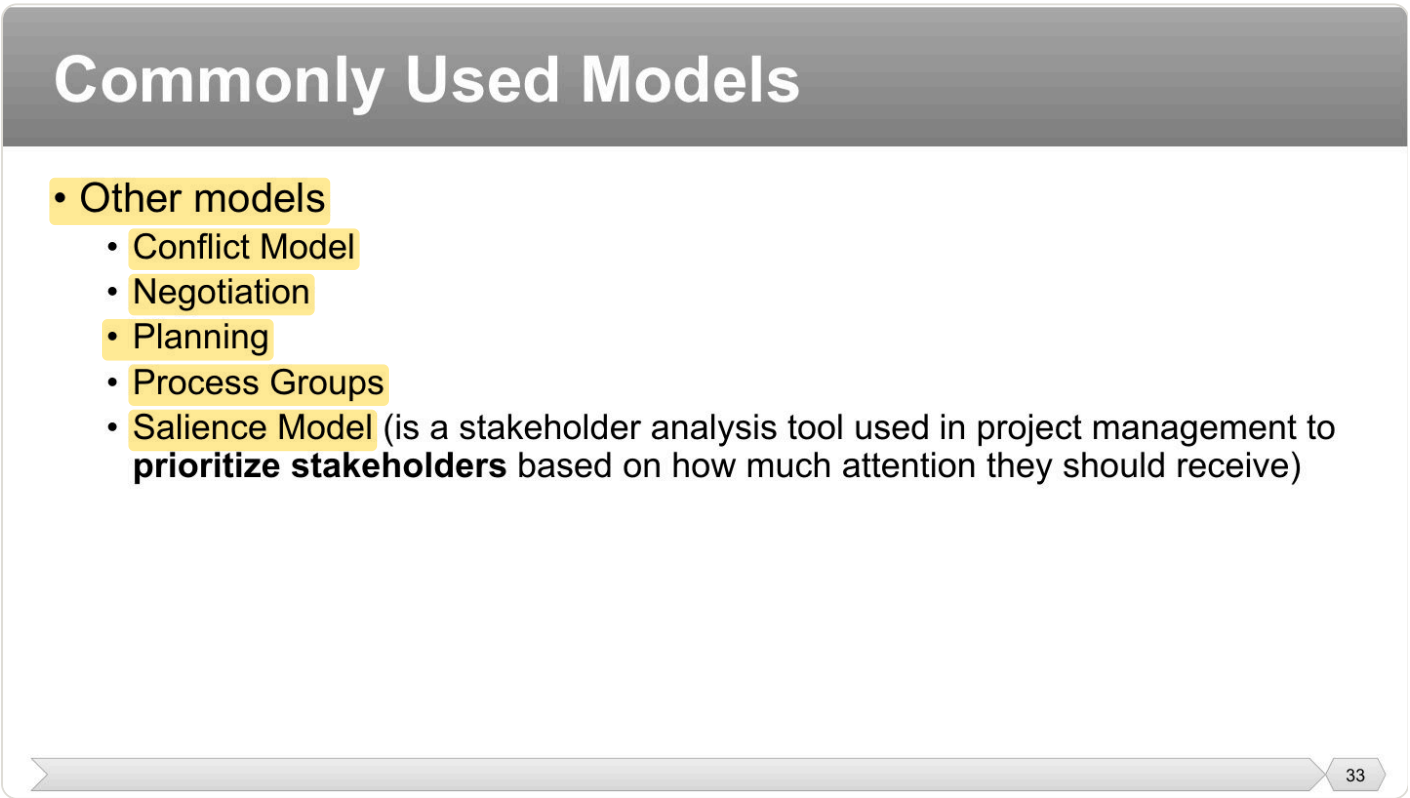
> Storming is when members compete for leadership and test group processes; norming comes after, when agreement is reached. (Also Midterm 251 V2 MCQ 7.)

B Norming

C Adjourning

D Performing

SLIDE 33 / 41 **Commonly Used Models**



NOTES

1. Other models: conflict (styles of handling conflict, Chapter 3), negotiation, planning, and process groups (initiating, planning, executing, monitoring and controlling, closing).
2. The **salience model** prioritizes stakeholders by **power, legitimacy and urgency** to decide how much attention each needs (Chapter 8).

EXAM ANGLE

Past exam · Final 251 · Q3.2 — Q1

How can project managers identify and prioritize stakeholders, and what strategies can they use to communicate effectively with each stakeholder group?

Answer: Identify stakeholders by brainstorming with the team, reading the charter and contracts, and interviewing key people, then record them in a stakeholder register. Prioritize them with a power/interest grid (or the salience model: power, legitimacy, urgency). Then tailor communication: high power and high interest → manage closely with interactive meetings; high power, low interest → keep satisfied with concise push reports; low power, high interest → keep informed with push updates or pull (portal); low/low → monitor. Chapter 8 develops stakeholder management.

SLIDE 34 / 41 Models by performance domain

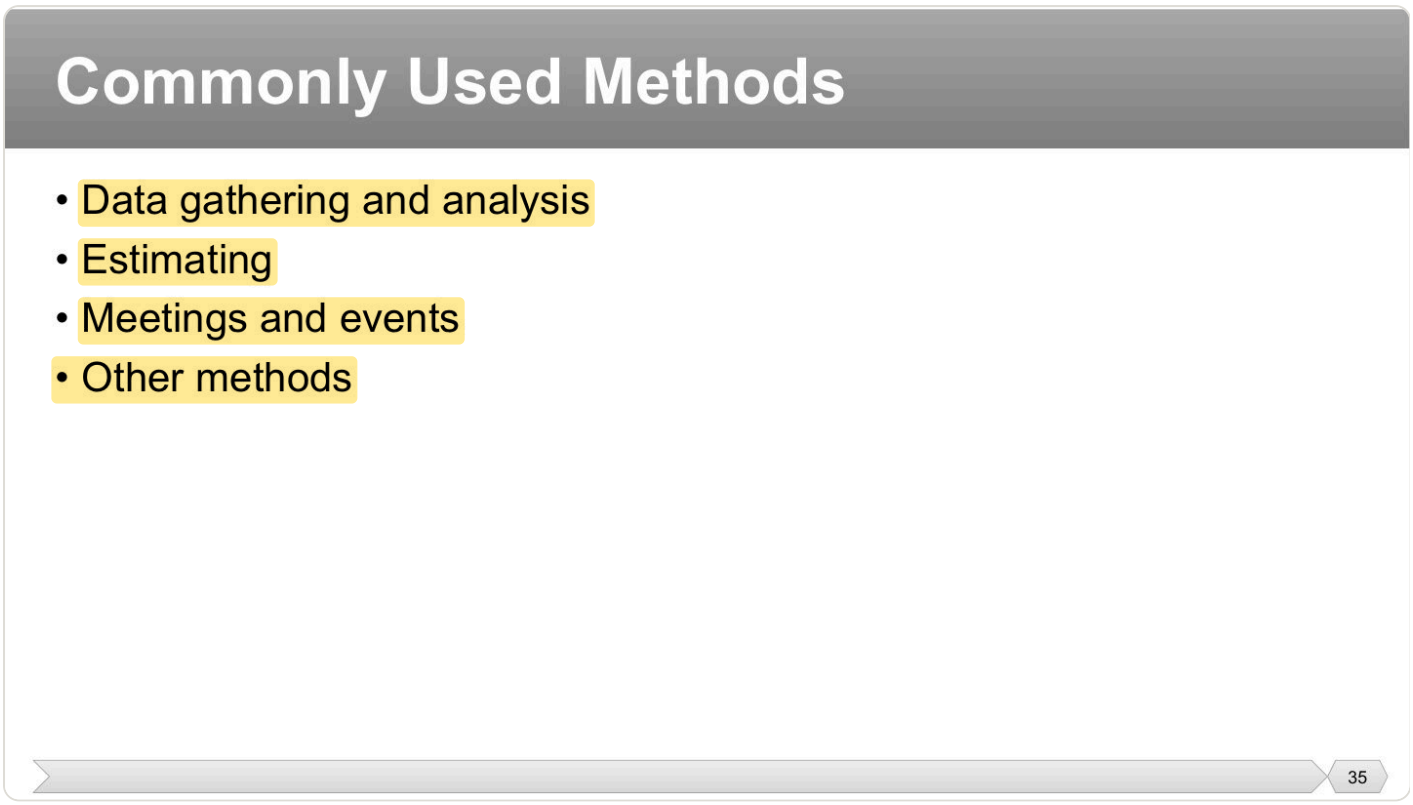
Model	Performance Domain							
	Team	Stakeholders	Dev Approach and Life Cycle	Planning	Project Work	Delivery	Measurement	Uncertainly
Situational Leadership Models:								
Situational Leadership® II	X				X			
OSCAR	X				X			
Communication Models:								
Cross-cultural communication	X	X		X	X			
Effectiveness of communication channels	X	X		X	X			
Gulf of execution and evaluation		X				X		
Motivation Models:								
Hygiene and motivation factors	X			X	X			
Intrinsic versus extrinsic motivation	X			X	X			
Theory of needs	X			X	X			
Theory X, Theory Y, and Theory Z	X			X	X			
Change Models:								
Managing Change in Organizations		X		X	X			
ADKAR®		X		X	X			
8-Step Process for Leading Change		X		X	X			
Transition		X		X	X			
Complexity Models:								
Cynefin framework			X	X	X	X		X
Stacey matrix			X	X	X	X		X
Project Team Development Models:								
Tuckman Ladder	X				X			
Drexler/Sibbet Team Performance	X				X			
Other Models:								
Conflict	X	X			X			
Negotiation		X		X	X	X		
Planning			X	X	X			
Process Groups				X	X	X	X	
Salience		X		X	X			

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NOTES

1. A reference table showing which performance domains each item is useful in (an X per domain). Don't memorize it; use it to pick a suitable tool when a scenario asks which method or artifact would help. Models: e.g. situational leadership and OSCAR help the team and project-work domains; the Stacey matrix helps choose the development approach and handle uncertainty.

 Commonly Used Methods



NOTES

1. Method families: **data gathering and analysis** (alternatives analysis, root-cause analysis, cost of quality, earned value, SWOT...).
2. **Estimating** (affinity grouping, function points, story points, Delphi...).
3. **Meetings and events** (kickoff, stand-up, retrospective, review, steering committee...), and other methods.

TIPS & TRICKS

- **DEMO**: Data gathering and analysis, Estimating, Meetings and events, Other methods.

SLIDE 36 / 41 Methods by performance domain

Method	Performance Domain							
	Team	Stakeholders	Dev Approach and Life Cycle	Planning	Project Work	Delivery	Measurement	Uncertainty
Data Gathering and Analysis Methods:								
Alternatives analysis				X	X	X		X
Assumptions and constraints analysis				X		X		X
Benchmarking						X	X	
Business justification analysis				X			X	
Payback period			X	X			X	
Internal rate of return				X			X	
Return on investment				X			X	
Net present value			X	X		X	X	
Cost-benefit ratio				X			X	
Check sheet						X	X	
Cost of quality				X		X	X	
Decision tree analysis				X				
Earned value analysis				X			X	
Expected monetary value				X				
Forecasting							X	
Influence diagram				X				
Life cycle assessment				X				
Make-or-buy analysis				X	X			
Probability and impact matrix				X				X
Process analysis				X	X	X	X	
Regression analysis				X			X	
Root cause analysis					X	X		
Sensitivity analysis				X	X	X		
Simulation				X			X	
Stakeholder analysis		X		X	X			
SWOT analysis				X				X
Trend analysis							X	
Value stream mapping				X	X	X		
Variance analysis							X	
What-if scenario analysis				X				X

NOTES

1 A reference table showing which performance domains each item is useful in (an X per domain). Don't memorize it; use it to pick a suitable tool when a scenario asks which method or artifact would help. Data methods: e.g. make-or-buy analysis serves planning and project work; probability-and-impact matrices serve uncertainty.

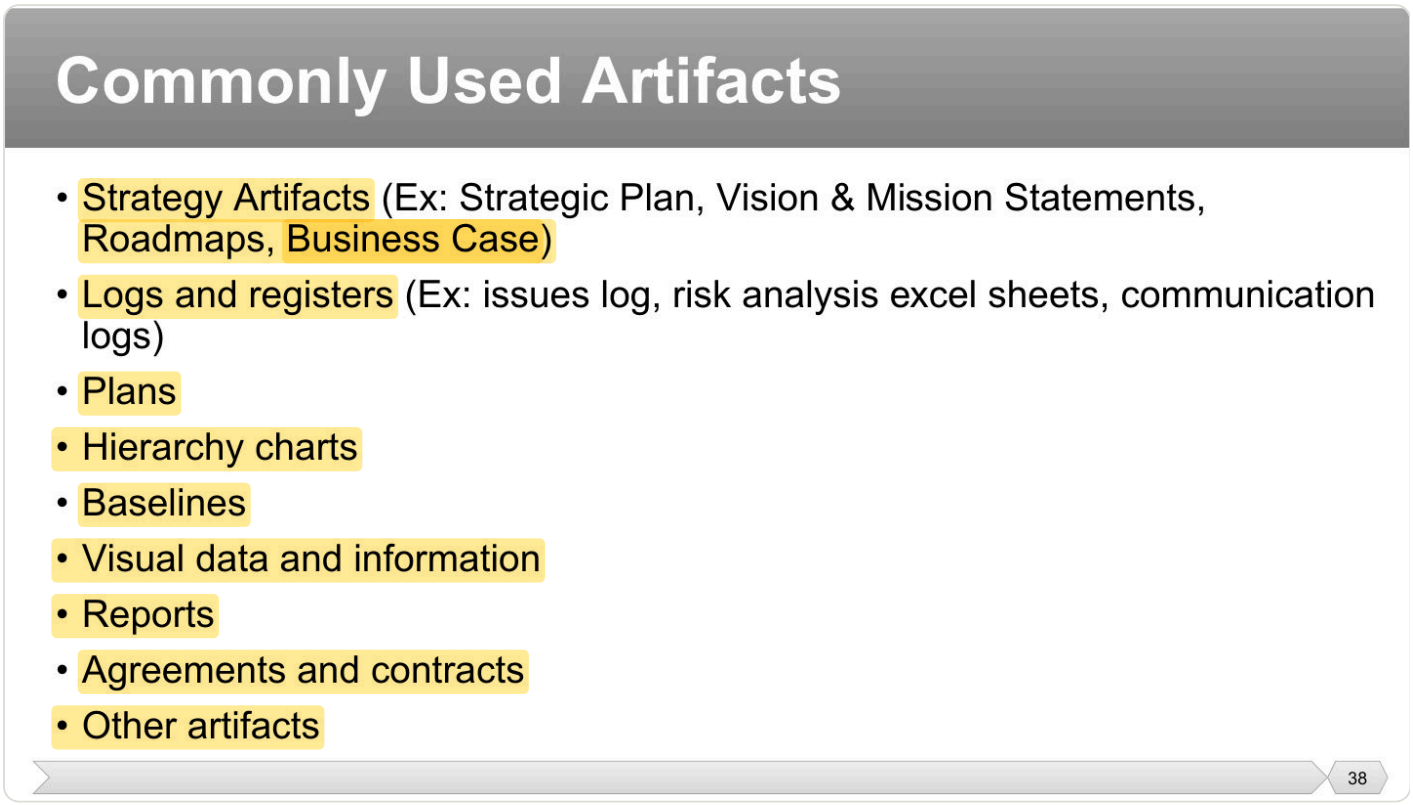
SLIDE 37 / 41 Methods by performance domain (cont.)

Method	Performance Domain							
	Team	Stakeholders	Dev Approach and Life Cycle	Planning	Project Work	Delivery	Measurement	Uncertainty
Estimating Methods:								
Affinity grouping				X				
Analogous estimating				X				
Function points				X				
Multipoint estimating				X				
Parametric estimating				X				
Relative estimating				X				
Single-point estimating				X				
Story point estimation				X				
Wideband Delphi				X				
Meeting and Event Methods:								
Backlog refinement		X		X	X	X		
Bidder conference		X		X	X			
Change control board					X	X		
Daily standup				X	X			
Iteration review		X			X	X		
Iteration planning		X		X	X	X		
Kickoff	X	X			X			
Lessons learned		X		X	X	X		
Planning				X				
Project closeout	X	X			X			
Project review		X			X	X	X	
Release planning		X		X				
Retrospective	X			X				
Risk review					X			X
Status					X		X	
Steering committee		X			X			
Other Methods:								
Impact mapping	X	X		X		X	X	
Modeling						X		
Net Promoter Score®		X					X	
Prioritization schema		X			X			
Timebox			X	X	X	X	X	

NOTES

1. A reference table showing which performance domains each item is useful in (an X per domain). Don't memorize it; use it to pick a suitable tool when a scenario asks which method or artifact would help. Estimating methods and meetings (e.g. retrospectives serve team and project work; risk reviews serve uncertainty).

SLIDE 38 / 41 Commonly Used Artifacts



NOTES

1. **Strategy artifacts**: strategic plan, vision and mission statements, roadmap, **business case**.
2. **Logs and registers**: issue log, risk register, communication log, stakeholder register, lessons-learned register.
3. Also **plans** (risk management plan, communications plan...), hierarchy charts (WBS, org chart), **baselines** (scope, schedule, cost), visual data (burn charts, Gantt charts), reports, **agreements and contracts**, and others.

TIPS & TRICKS

- Sentence, in slide order: **S**tudents **L**ove **P**izza, **H**ot **B**urgers, **V**anilla **R**olls **A**nd **O**reos. Strategy, Logs and registers, Plans, Hierarchy charts, Baselines, Visual data, Reports, Agreements, Other.

EXAM ANGLE

Prediction

Q1

A business case belongs to which category of artifacts?

A Reports

B Logs and registers

C **Strategy artifacts**

D Baselines

SLIDE 39 / 41 **Artifacts by performance domain**

Artifact	Performance Domain							
	Team	Stakeholders	Dev Approach and Life Cycle	Planning	Project Work	Delivery	Measurement	Uncertainty
Strategy Artifacts:								
Business case		X		X				
Project brief		X		X				
Project charter		X		X				
Project vision statement		X		X				
Roadmap		X	X	X				
Log and Register Artifacts:								
Assumption log				X	X	X		X
Backlog				X	X	X		
Change log					X	X		
Issue log					X			
Lessons learned register					X			
Risk-adjusted backlog				X				X
Risk register				X	X	X		X
Stakeholder register		X		X				
Plan Artifacts:								
Change control plan				X	X	X		
Communications management plan		X		X	X			
Cost management plan				X				
Iteration plan				X				
Procurement management plan				X	X			
Project management plan		X		X	X			
Quality management plan				X	X	X		
Release plan				X		X		
Requirements management plan				X		X		
Resource management plan				X	X			
Risk management plan				X	X			X
Scope management plan				X		X		

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NOTES

1 A reference table showing which performance domains each item is useful in (an X per domain). Don’t memorize it; use it to pick a suitable tool when a scenario asks which method or artifact would help. Artifacts: e.g. the business case and charter serve stakeholders and planning; the risk register serves planning, project work, delivery and uncertainty.

SLIDE 40 / 41 Artifacts by performance domain (cont.)

Artifact	Performance Domain							
	Team	Stakeholders	Dev Approach and Life Cycle	Planning	Project Work	Delivery	Measurement	Uncertainty
Schedule management plan				X	X	X		
Stakeholder engagement plan		X		X				
Test plan				X	X	X	X	
Hierarchy Chart Artifacts:								
Organizational breakdown structure	X	X		X				
Product breakdown structure				X		X		
Resource breakdown structure	X			X	X		X	
Risk breakdown structure					X			X
Work breakdown structure				X		X	X	
Baseline Artifacts:								
Budget				X	X		X	
Milestone schedule			X	X	X		X	
Performance measurement baseline				X	X	X	X	
Project schedule				X	X		X	
Scope baseline				X	X	X	X	
Visual Data and Information Artifacts:								
Affinity diagram				X	X			
Burn chart				X		X	X	
Cause-and-effect diagram					X	X		X
Cycle time chart						X	X	
Cumulative flow diagram						X	X	
Dashboard					X		X	
Flow chart				X	X	X		
Gantt chart				X	X		X	
Histogram							X	
Information radiator					X		X	
Lead time chart						X	X	
Prioritization matrix		X			X	X		

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NOTES

1 A reference table showing which performance domains each item is useful in (an X per domain). Don't memorize it; use it to pick a suitable tool when a scenario asks which method or artifact would help. Hierarchy charts, baselines and visual data (e.g. burn charts serve measurement).

SLIDE 41 / 41 Artifacts by performance domain (cont.)

Artifact	Performance Domain							
	Team	Stakeholders	Dev Approach and Life Cycle	Planning	Project Work	Delivery	Measurement	Uncertainty
Project schedule network diagram				X	X			
Requirements traceability matrix				X		X	X	
Responsibility assignment matrix				X	X			
Scatter diagram					X	X	X	
S-curve				X			X	
Stakeholder engagement assessment matrix		X		X	X			
Story map				X		X		
Throughput chart						X	X	
Use case				X		X		
Value stream map					X	X	X	
Velocity chart						X	X	
Report Artifacts:								
Quality report					X	X	X	
Risk report					X			X
Status report					X			
Agreements and Contracts:								
Fixed-price		X		X	X	X	X	X
Cost-reimbursable		X		X	X	X	X	X
Time and materials		X		X	X	X	X	X
Indefinite time indefinite quantity (IDIQ)		X		X	X	X	X	X
Other agreements		X		X	X	X	X	X
Other Artifacts:								
Activity list	X	X		X	X			
Bid documents		X		X	X			
Metrics				X		X	X	
Project calendars	X			X	X			
Requirements documentation		X		X		X	X	
Project team charter	X				X			
User story		X		X		X		

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NOTES

1. A reference table showing which performance domains each item is useful in (an X per domain). Don’t memorize it; use it to pick a suitable tool when a scenario asks which method or artifact would help. Reports, agreements and contracts.

Original slides from the SE423 lecture deck, with study notes. Check exact wording against the original deck.
Notes and questions are study aids. Made by Shoug Alomran.